Hertie School

Master's Thesis

Roles that Sustain Us: Volunteering, Retirement, and Health Trajectories in an Ageing Germany

Samuel Perez Restrepo

Master of Public Policy

2026

Word Count: 12,144

Supervised by Prof. Dr. Michaela Kreyenfeld

Acknowledgment

I am deeply grateful to the many people who supported me throughout this thesis process. First, I want to thank my family and friends, whose encouragement, patience, and steady interest in my progress sustained me from start to finish.

I am also sincerely thankful to the Better Politics Foundation and the DIW team, especially Ximena D. and Daniel G., for their suggestions, encouragement, and support from the very beginning of the project. I am equally grateful to CIVIX Chile, particularly to Alexia V., for giving me the opportunity to engage directly with civic engagement work, which helped shape both my academic interests and this thesis.

My greatest thanks go to Prof. Dr. Michaela Kreyenfeld for her supervision. Her guidance, rigor, and generosity strengthened this project at every stage. Working with her allowed me to deepen and refine my analytical approach and to develop a thesis that is both academically meaningful and closely connected to the pressing issues of our time.

Table of Contents

Executive Summary

Retirement is a major life-course transition that can reshape daily routines, social roles, and well-being. This thesis examines whether adults in Germany who volunteered before retirement experienced different self-rated health trajectories around the retirement transition than those who did not. Using longitudinal data from the German Socio-Economic Panel and an event-time design, the analysis compares health from two years before to five years after retirement, with volunteering measured shortly before the transition. The results show a clear descriptive pattern: individuals who volunteer pre-retirement report better self-rated health around retirement than non-volunteers.

At the same time, the multivariate and fixed-effects models suggest a more cautious interpretation. Pre-retirement volunteers appear to enter retirement in better health, but there is limited evidence that volunteering substantially alters short-run within-person health trajectories after retirement. In other words, volunteering is more strongly associated with a health advantage at the transition than with a distinct post-retirement health path. These findings matter for both theory and policy. They support the view that civic engagement is part of the broader social infrastructure of healthy aging, but they do not justify treating volunteering as a stand-alone health intervention. In the German context, the implication is that policies should not only encourage volunteering among older adults but also reduce the social and organizational barriers that make civic engagement easier for some groups than for others.

1. Introduction

“We only have what we give.”

— Isabel Allende

Civic engagement is part of every functional society, since it involves the individual and collective actions people take to identify and address issues of public concern (American Psychological Association, 2006). It has been studied for decades, and although there are persistent disputes about its boundaries, its importance is evident: civic engagement sustains democratic life, strengthens communities, and shapes how people relate to one another (Adler & Goggin, 2005). My motivation for studying this topic comes from a mix of academic interest, professional exposure, and personal experience. Growing up with parents working in the health sector, I often heard stories about how illness can weaken not only the body but also the social connections that keep people grounded. Illness can exclude individuals from their close circles, their neighborhoods, and eventually from broader community life, forming a cycle in which social withdrawal reinforces health problems. These observations correspond to the larger struggles many societies face today: rising polarization, declining trust, and increasing sensations of loneliness and disconnection.

At the same time, concerns about both physical and mental health have become more prominent in many countries. Even though awareness of mental health has improved over the last two decades, indicators such as depressive symptoms, anxiety, and suicide rates remain high. Much of this decline has been linked to social determinants of health, including income, gender, education, housing, and discrimination (Marmot et al., 2008). Germany faces many of the same pressures. Despite having a strong welfare state and a strong health system, it faces the same socio-political pressures seen elsewhere. These issues make the intersection between civic engagement and health particularly relevant. Germany’s civic infrastructure, especially its network of more than 600,000 associations (Schubert et al., 2022), is widely praised as an example for other democracies. These organizations reflect a long tradition of participation and cooperative problem-solving, but they are exposed to social fragmentation or declining participation (Alscher, 2024). Understanding how civic engagement connects to individual well-being in this context is very timely and necessary.

Studying civic engagement and health together is important because their relationship is complex and multidimensional. Engagement can take many forms, such as volunteering, activism, and membership in associations, each of which may have different ramifications for well-being. Some research suggests that civic engagement can reduce stress, elevate mood, increase physical activity, strengthen social networks, and provide a sense of meaning (Piliavin & Siegl, 2007; Thoits & Hewitt, 2001). Conceptual models from developmental psychology, such as positive youth development (Damon, 2004; Lerner et al., 2005) and sociopolitical development (Watts et al., 1999, 2011), help explain why engagement might influence well-being, pointing to mechanisms such as social connectedness, empowerment, and behavior management (Ballard et al., 2019). Yet despite growing interest in these pathways, there is still limited evidence on how civic engagement relates to within-person changes in health over time, especially in Germany.

I address these gaps by focusing on a pivotal life-course transition: retirement. Retirement reshapes daily routines, social ties, and personal identity, and its health effects can vary widely from person to person. My central question is whether volunteering shortly before retirement is associated with distinct self-rated health trajectories during this stage. Instead of seeing volunteering as a straightforward cause of health outcomes, I treat it as a pre-existing role that might cushion or intensify the health consequences of retirement.

The policy relevance is clear. If volunteering is associated with entering retirement in better health, it may be part of broader healthy-aging strategies, especially in settings where traditional interventions have limited reach. Strengthening civic infrastructure, supporting volunteering programs, or reducing barriers to participation could have indirect benefits for well-being. Conversely, if the relationship is weak or varies across groups, policymakers need to understand these nuances before relying on civic engagement as a health-promoting strategy. In a period distinguished by social fragmentation, declining trust, and rising mental health concerns, evidence-based knowledge of the civic–health connection is valuable for designing interventions that support both democratic life and population well-being.

To help the reader navigate my approach, I first review the conceptual foundations and prior evidence linking civic engagement to health, then lay out my model and core hypotheses. After presenting my data and defining key variables, I walk through the results, including descriptive patterns, longitudinal models, and robustness checks. Finally, I discuss the implications of these findings for policy and theory and highlight directions for future research.

2. Theory and Prior Evidence

2.1 Conceptual Foundations: Roles, Continuity, and Civic Engagement

Retirement marks one of the most crucial phases in life, altering people's social roles, routines, and identities. In the classic role theory of retirement, the phenomenon is considered a role exit, in which people leave their major workplace roles that structured their time, facilitated social interactions, and enhanced their sense of purpose and personal identity (Thoits, 2012). Such an exit creates role voids that increase vulnerability to negative health and psychological states if there is no other role to take its place and ensure continuity in people's everyday life routines. The continuity theory similarly states that individuals who continue engaging in valuable activities and maintain roles after retirement will experience more comfortable transitions and enjoy better health. Economic theories of retirement also support this point of view. Bloom et al. (2007) define retirement as the process of reallocation of time and psychological resources.

Under this approach, volunteering proves especially important as an alternative role. Volunteering offers routine and structure, as well as social contact and a sense of purpose, all of which are psychological constructs similar to those offered by employment. Kaskie et al. (2008) explicitly consider civic engagement a retirement role, suggesting that volunteering serves as an adequate substitute for fulfilling employment-related roles and promoting successful aging. German/European empirical research confirms this substitution approach to the role concept: Erlinghagen (2010) demonstrates that prior experience of volunteering is a very strong predictor of future volunteering after retirement, whereas Bogaard et al. (2014) reveal that retirement may change the patterns of civic engagement, making certain people more active volunteers than previously. Thus, this role substitution hypothesis provides theoretical grounds for analyzing volunteering in the pre-retirement stage, defined as participation in the year immediately preceding retirement, based on the latest non-missing observation within the one- to three-year interval prior to retirement.

The link between civic participation and health has been studied using several theories, all of which highlight the various ways civic participation may affect one's health and well-being (Nelson et al., 2019). Social capital theory examines how social networks, reciprocity, and trust facilitate access to social resources that promote good mental and physical health (Putnam, 2000). Participation in volunteerism can build such networks and provide individuals with greater opportunities to access social support, group identity, and collective efficacy. In psychology, volunteering can play an important role in society, thereby promoting individuals' self-respect and self-efficacy. At the same time, stress-buffering models posit that it helps reduce stress by providing instrumental, informational, and emotional support (Cohen & Wills, 1985). From the developmental perspective, contributing to society, feeling part of a community, and learning new skills can enhance one's well-being throughout one's lifetime (Hershberg et al., 2015). All of these mechanisms contribute to the potential ability of volunteering to prevent the adverse health effects of retiring from active work.

The rewards that may be obtained from volunteering are likely to differ depending on one's socioeconomic position. Indeed, resource substitution theory implies that the marginal utility of acquiring additional resources for one's psychological and social well-being is higher for people who are less privileged in structural resources. As concerns retired people, those in the lowest SES are expected to have fewer options for social integration, a sense of purpose, and time structuring. Indeed, the results of a recent longitudinal study support this reasoning, showing that the health benefits of volunteering are greater for poorer individuals (Hobfoll, 2011; Van Willigen, 2000).

The influence of volunteering might also vary for men and women depending on gender-specific life-course patterns. While men’s career paths have always been relatively uninterrupted and their identities have been strongly bound up with work, women's lives have been marked by career interruptions and periods of part-time work (Moen, 2001; Quick & Moen, 1998). For men, retirement is a more decisive phase-out of a crucial sphere of life, increasing the importance of alternative roles such as volunteering (Van Ingen & Wilson, 2017). For women combining work, caregiving, and civic engagement, volunteering can also serve as a rare role within the non-domestic sphere (Moen & Flood, 2013; Taniguchi, 2006). The gender differences in the influence of volunteering before retirement should thus not be attributed to different mechanisms, but rather to different meanings attached to retirement and different life courses preceding it.

2.2 Prior Research

Literature on retirement, volunteering, and health has expanded dramatically over the last few decades. The three main streams of literature, which are most directly relevant to this thesis, retirement and health, volunteering and health among older adults, and retirement and volunteering, have advanced relatively independently from each other. This subsection provides an overview of these three streams.

The core difficulty in examining the effect of retirement on health outcomes is endogeneity, because people with health problems are likely to retire earlier. Certain methodological designs help reduce the risk of endogeneity; for example, recent research uses natural experiments based on pension eligibility criteria.

In Germany, Eibich (2015) offers some of the strongest causal evidence available. Using a regression discontinuity design based on statutory retirement ages, he showed that retirement positively affects self-perceived health status due to reduced work stress, increased sleep, and increased physical activity. Likewise, positive outcomes were also reported in other countries. In England, Rose (2020) employed a similar RDD framework and found that retirement positively affects well-being and self-perceived health.

While there is strong evidence of the positive impact of retirement on health, Heller-Sahlgren (2012) reported a negative medium- to long-run effect of retirement on self-perceived health, using SHARE data and spouses as instrumental variables.

Additionally, Hanemann (2017) analyzed harmonized data from the United States and Europe and demonstrated that retirement enhances physical health but may worsen cognitive health. Moreover, retirement had greater benefits for people working in stressful occupations.

These findings illustrate the differential effects of retirement on health, depending on job strain, socioeconomic background, and health prior to retirement. As a result, this explains the motivation for this study and the search for moderators of the effects of retirement on health.

Numerous academic articles have been published discussing the link between volunteering and better health outcomes in older age groups. For instance, according to Morrow-Howell et al. (2003), there is a positive association between volunteering, increased well-being and self-rated health, independent of initial conditions. The trend is also evident in recent evidence collected from European panel data samples. According to de Wit, Qu, and Bekkers (2022), a thorough statistical analysis of data from six panel surveys with 950,000 participants revealed a statistically insignificant correlation between increases in volunteering and gains in self-rated health, especially among people aged 60 years and above and those who were unhealthy.

Nevertheless, most recent studies emphasize the reciprocity of the connection under discussion. For example, Węziak-Białowolska et al. (2024), using the SHARE dataset, find that although there is a positive relationship between volunteering and emotional well-being, as well as fewer difficulties in performing daily tasks, high levels of well-being and social integration tend to predict future volunteering. In Germany, similar results were achieved by Meier and Stutzer (2008) and Lühr (2025). Therefore, it can be assumed that much of the connection in question may be attributable to selection effects.

The selection issue at hand is central to the current analysis. Unlike other studies that view volunteering as a causal predictor of health, this study perceives prior volunteering as a moderating variable in the process of retirement, based on empirical findings showing that volunteering is characterized by enduring role commitments and can influence how people cope with life transitions (Greenfield & Marks, 2004; Pavlova & Silbereisen, 2012).

Direct research into the relationship between retirement and volunteering remains scarce. Most researchers focus on the effect of retirement on volunteering or use volunteering as a causal predictor of health, ignoring its potential role in moderating retirement's impact on health.

Volunteering becomes more common in retirement. According to Hämäläinen et al. (2024), drawing on SHARE data from 19 nations, the onset of retirement is associated with higher volunteering rates, particularly among educated and healthier individuals. Eibich et al. (2022) validated the causality of this connection by employing pension eligibility as an instrumental variable. Previous German evidence from Erlinghagen (2010) also revealed that previous volunteering is a key predictor of sustained volunteering in retirement.

A limited number of studies have examined the link between volunteering and well-being during or after retirement. Longitudinal evidence consistently shows that volunteering is positively associated with well-being and reduced depressive symptoms in older adults (Li & Ferraro, 2005; Morrow-Howell et al., 2003; Weziak-Bialowolska et al., 2024). Conversely, retiring from a volunteering role is associated with poorer health and well-being outcomes (Russell et al., 2023). However, no study has yet investigated whether pre-retirement volunteering moderates the impact of the retirement transition itself on health trajectories over time.

2.3 Evidence Linking Civic Engagement and Mental Health

There is considerable evidence that volunteering is positively associated with mental health benefits for older persons. Volunteering seems to be a protective factor in terms of depressive symptoms, life satisfaction, and loneliness. Studies show that people who volunteer experience a delayed increase in depressive symptoms and more stable well-being over time (Li & Ferraro, 2006; Lum & Lightfoot, 2005). This shows that volunteering protects people against the loss of psychological resources associated with aging by preserving social roles, a sense of purpose, and routine.

Loneliness becomes especially important for retired people because retirement usually leads to a decline in daily social contact. Panel surveys conducted in Europe have shown a negative relationship between volunteering and loneliness among older adults, though this is not always the case, depending on gender and socioeconomic factors (Hansen et al., 2018). Recent studies using HRS data in the United States have also shown that civic engagement reduces loneliness among older adults (Hayes et al., 2025).

Furthermore, longitudinal data acquired from within-person analyses suggest that the effects of volunteering on psychological well-being are restricted. That is, in the study by Lühr (2025), which examined changes in individuals from German and British samples over an extended period, volunteering does not boost either life satisfaction or positive mood in adults when controlling for stable individual characteristics. Additionally, it was found that political activism might lead to a growing sense of loneliness and reduced life satisfaction among certain groups, illustrating that different forms of civic engagement may operate in distinct ways.

However, various civic activities such as political participation can help middle-aged people feel empowered and develop generativity (Keyes & Ryff, 1998; Serrat et al., 2017); nevertheless, the effects of these engagements on individuals' psychological states are less certain than the benefits of volunteering. Indeed, within-person analyses conducted longitudinally, have found that both political and non-political civic engagement have a comparable effect on well-being (Lühr et al., 2022a). Inasmuch as political engagement cannot offer individuals the same context as volunteering, this activity should be regarded as a complementary lens or an independent study.

2.4 Evidence Linking Civic Engagement and Physical Health

Much of the literature on civic engagement and health among older adults centers on self-rated health, functional difficulties, and mortality. There is substantial research linking volunteering to better SRH, wherein those who volunteer have higher levels of general health and fewer functional difficulties. Several early studies have linked volunteering to better physical functioning and lower mortality risk (Okun et al., 2013; Piliavin & Siegl, 2007). An extensive outcome-wide study employing HRS data from the United States established an association between high levels of volunteering, defined as 100 or more hours of volunteering per year, and lower mortality risks, fewer physical functioning limitations, and better health behavior (E. S. Kim et al., 2020; S. Kim et al., 2025). Recently, an analysis of HRS data revealed that the relationship between volunteering and mortality risk was mediated by physical and social variables, including contact and helping friends, neighbors, and family members (Nakamura et al., 2025).

Volunteering can impact physical health through exercise, healthy habits, and better stress management (Anderson et al., 2014). Various types of volunteer activities, including light exercise, contribute to mobility and heart health, while social integration facilitates healthy behavior (Brown et al., 2017; Schreier et al., 2013). Among older people, being an active volunteer has been associated with slower deterioration in physical function and a lower risk of chronic illnesses (Han et al., 2018; S. Kim et al., 2023; Lum & Lightfoot, 2005; Sneed & Cohen, 2013).

A possible mechanism by which the two phenomena are connected is that volunteering influences health behaviors directly. Older volunteers tend to engage in less risky health behaviors, such as smoking, heavy alcohol use, and leading a sedentary life, while also seeking more preventive health care, such as screening and vaccinations, compared to their counterparts who do not volunteer (Harris & Thoresen, 2005; E. S. Kim & Konrath, 2016). Evidence of a causal effect comes from the Experience Corps intervention study, where elderly participants volunteered in schools, and the randomized experiment led to increased physical activity and cardiovascular benefits in terms of inflammatory markers and cholesterol levels (Fried et al., 2004). Nevertheless, caution should be exercised in considering the extent of health behaviors that volunteering may influence. Outcome-wide longitudinal analysis based on HRS data has shown that, although high-intensity volunteering is associated with a 12% increase in physical activity, there were no observed connections between it and binge drinking, smoking, or insomnia (E. S. Kim et al., 2020a). Therefore, volunteering influences physical health only in terms of physical activity and preventive health care, but not substance abuse.

However, causal evidence still lacks in number. The majority of studies use cross-sectional design and cannot exclude reverse causality, according to which only healthy persons are more likely to participate in volunteering (Li & Ferraro, 2005; Papa et al., 2019). The selection effect was confirmed by longitudinal panel data in Europe, according to which poor health condition, in particular, functional limitations, decreases the chance of volunteering (Papa et al., 2019). Using within-persons approach to distinguish changes from selection leads to the fact that although the effect becomes weaker, the correlation between volunteering and SRH stays statistically significant and minor (de Wit et al., 2022). This finding contradicts the results obtained from meta-analysis of mortality risk and mortality rate (Okun et al., 2013), but such inconsistency can be explained by different time span and higher sensitivity to changes.

On the other hand, other types of civic engagement, such as political engagement, have shown a lower level of correlation and consistency with better physical well-being. Although some studies have linked political engagement to better SRH, the correlations remain lower and less significant than those found for volunteering, since political engagement does not involve regular behavioral regulation. According to Nascimento et al. (2025), research on European samples comprising 14,700 people aged 50 years or more found that individuals with better SRH tend to participate in political activities, with physical activity playing an important role in mediating this relationship. Here, the correlation runs in the opposite direction: individuals who are healthy engage in political activity more often than those who are not, suggesting that political activity may not actually cause health improvements, but that healthier persons tend to engage more in politics. Lastly, according to Piumatti et al. (2018), political interest mediated the relationship between social trust and self-rated health among Italian older adults, but the effect was indirect and weak.

All things considered, it can be said from the literature that volunteerism is the type of civic participation most strongly correlated with health in older adulthood, despite the fact that some of this effect is probably due to selection. Clearly, the literature highlights the importance of analyzing how volunteerism prior to retirement influences health outcomes, rather than viewing civic engagement itself as causal.

2.5 Gaps in the Literature and How This Thesis Addresses Them

In addition, most studies assessing the connection between civic engagement and well-being of older individuals are cross-sectional, thus not allowing one to identify causal directions or relations between the variables (Filges et al., 2020; Jenkinson et al., 2013); even when employing longitudinal designs, within-person effects are measured far less often compared to between-person effects. As already seen in earlier SOEP research by Meier & Stutzer (2008), the relationship between civic engagement and well-being is largely caused by self-selection effects. In more recent panel data analyses, it was also shown that most beneficial aspects of civic engagement on mental well-being are negligible if one controls for unobserved individual heterogeneity (Lühr et al., 2022b, 2022a). Cross-country comparisons of Europe do provide results showing some associations between civic engagement and well-being (Pavlova & Lühr, 2023; Vega-Tinoco et al., 2022), yet causal evidence based on German data seems lacking, considering the specificities of German civic infrastructure (Evers, 2019).

Another important issue is the problem of mechanisms. Although concepts such as loneliness, social connectivity, and perceived control have been proposed as potential mediators in the relationship between civic engagement and well-being, few studies have explored these links using longitudinal or intra-individual approaches. In the case of loneliness, for instance, this concept was shown to be an appropriate mediator of the process of health decline as people age (Cacioppo & Cacioppo, 2014). Nevertheless, longitudinal research investigating the influence of civic engagement on loneliness around life events is rather scarce. However, the proposed pathways can explain interest in studying volunteer roles.

The third gap exists at the juncture where retirement and community engagement meet. Volunteering has largely been examined in terms of its impact on retirees and the broader context of changes in social roles in later years, rather than its pre-retirement impacts on how individuals transition into retirement (Heaven et al., 2013; Lehane & Scarlett, 2024; Russell et al., 2023). Very little is known about whether participation in volunteer work before retirement affects well-being or health after retirement. Despite evidence showing that post-retirement volunteering improves psychological well-being and that interventions involving social roles during the transition to retirement improve well-being and health, research on this topic remains limited.

A final area that remains unexplored concerns the German setting. The strong tradition of Vereinswesen (associational life), the established civic infrastructure, and the unique welfare state create distinct volunteering and retirement trends in Germany compared to Anglo-American settings (Evers, 2019). However, despite extensive longitudinal data, the interplay between volunteering and retirement, and their relationship to health trajectories in the German context, remains under-researched.

By investigating this research question, the thesis aims to bridge some of these gaps through the use of longitudinal data collected in the German Socio-Economic Panel on whether engaging in voluntary work prior to retirement, as measured by the last available observation within the one- to three-year period preceding retirement, is linked to distinct self-assessed health trajectories during the time of retirement in Germany. The contribution is narrower and more focused than the neighboring literature. This thesis does not analyze the quasi-experimental effect of retirement, unlike research on the causal effects of retirement using pension-age discontinuities, such as Eibich (2015) and Rose (2020). Unlike research papers that examine whether volunteering activities precede health outcomes, as in de Wit et al. (2022), this paper does not examine whether a change in the level of volunteering activity results in health-related outcomes. This research also diverges from the literature demonstrating a positive relationship between retirement and volunteering, such as Hämäläinen et al. (2024) and Eibich et al. (2022), as the former examines the effect of retirement on volunteering, whereas the latter treats volunteering as a result of retirement. This research examines whether volunteering before retirement affects the trajectory of health outcomes during retirement transitions in Germany.

3. Research Question and Hypotheses

3.1 Research Question

This thesis asks how self-rated health evolves around the observed transition to retirement among adults aged 50 to 70 in Germany who experience an observed retirement transition during the observation window, and whether these trajectories differ by pre-retirement volunteering.

In particular, pre-retirement volunteering is defined as the most recent non-missing observation in the one- to three-year period before the transition, because it addresses concerns about temporal order and accounts for the data collection process and the civic engagement module's time frame.

Reverse causation is still an important issue to consider, since healthier people tend to volunteer and remain engaged with their community. The use of volunteering data collected prior to retirement, based on the most recent observation within the one- to three-year period, allows better consideration of the temporal aspect but will not entirely address problems of reverse causation or selection. Income can also be considered endogenous, since the retirement decision has a direct mechanical effect on income.

3.2 Conceptual Diagram and Causal Pathways

Figure 1 summarizes the conceptual pathways linking retirement, pre-retirement volunteering, and self-rated health, as well as the background factors that may shape both engagement and health.

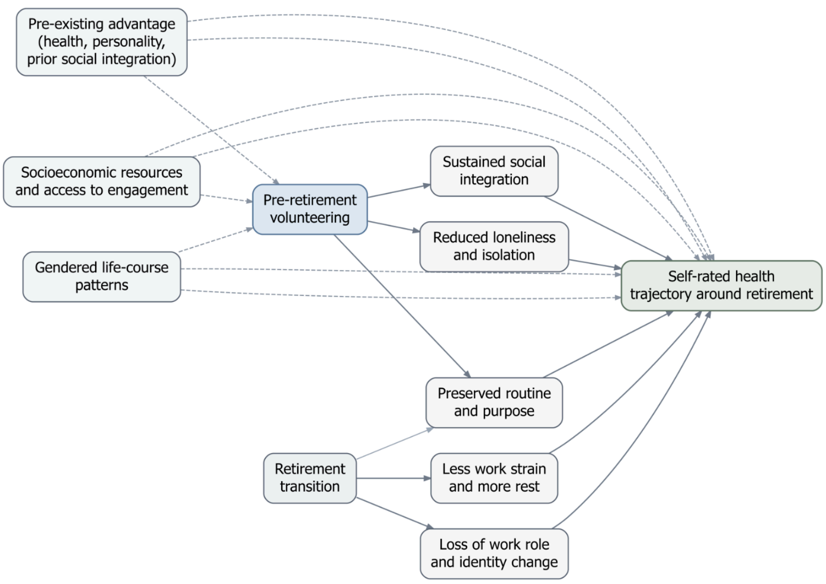


Figure 1. Conceptual pathways linking retirement, pre-retirement volunteering, and self-rated health.

Some theories explain that changes in the nature of activities, social relationships, and one's identity that accompany retirement, natural of exiting work, might adversely affect an individual's well-being by causing the loss of meaning and integration in society. Empirical studies, however, show that retirement can positively affect people's health by removing the pressures of employment and allowing for more sleep. Since both positive and negative outcomes are likely in retirement, the critical issue is who achieves which outcome and at what point in their life.

The role substitution hypothesis implies that volunteering before retirement influences the kind of adjustment experienced after retirement. Volunteering before retirement indicates established patterns of civic behavior linked to healthy transitions around retirement, including continued social integration, a sense of purpose and routine, and reduced loneliness.

Given resource-substitution approaches, the advantages associated with volunteer work are unlikely to be similar across all segments of society. Individuals from lower socioeconomic backgrounds are expected to gain more from volunteerism because their lives lack organization and meaning, which they can only find elsewhere, making volunteering more significant for them during retirement. Life-course gender differences will also play an essential role. The career histories of males are relatively uninterrupted compared with those of females, whose careers are frequently disrupted by child-rearing responsibilities and part-time employment. Therefore, retirement will signify role exit for the former group and a means of integrating volunteering into their everyday routines for the latter. This is the rationale for the additional hypothesis that the relationship between health trajectories and volunteering varies by socioeconomic status and gender.

3.3 Hypotheses

H1. Retirement and Health

Retirement is expected to be associated with a short-term improvement in self-rated health around the transition, followed by a gradual deterioration in later post-retirement years. This reflects evidence that retirement can reduce work-related stress and improve rest and health behaviors, even as aging continues to exert downward pressure on health.

H2. Moderation by Prior Volunteering

Individuals with pre-retirement volunteering are expected to exhibit more favorable self-rated health trajectories around retirement than those without prior volunteering, suggesting that volunteering may provide structure, social integration, and continuity during the transition.

4. Data and Variables

4.1 Data Source

Data for this thesis come from the German Socio-Economic Panel (SOEP, v41), a nationally representative panel survey conducted annually in Germany since 1984 (DIW, 2026; Goebel et al., 2019). SOEP provides rich data on socio-economic indicators, labor market activities, family relations, physical and mental well-being, and civic engagement; hence, it can be regarded as an ideal dataset for studying transitions such as retirement. The panel character of SOEP enables longitudinal studies that capture individual changes over time, including changes in self-rated health before and after retirement.

The analytical window will cover the years 2003 to 2023. Such a timeframe was chosen to ensure access to harmonized data on self-rated health, retirement status, and volunteering behavior. Not all SOEP respondents constitute the analytical sample for this paper. Only the people who have undergone their first transition to retirement within the observation window will be considered in the sample. This choice is determined by the very nature of the research question posed.

The final analytical sample is further restricted to observations within an event-time window spanning two years prior to retirement through five years following retirement. After applying all sample restrictions, the final analytical sample based on individual-level data includes 8,314 people, of whom 4,228 are female (50.9%), and 4,086 are male (49.1%). The final individual-year panel comprises 52,135 individual-year observations with event times ranging from -2 to +5.

The selected analytical sample will be inherently selective, as only those who experienced an observed retirement transition and had a valid measure of volunteering prior to retirement could be included in this study. This point becomes especially important with regard to gender since women within these cohorts may be underrepresented among those people who experience an observed retirement transition during the observation period.

4.2 Civic Engagement Measures

The major moderator used in this thesis is volunteering before retirement. The measure of volunteering uses harmonized SOEP items from the pli0096 family to assess the frequency with which respondents engage in volunteer or charitable work with clubs or organizations. The use of items involves pli0096_h, pli0096_v1, pli0096_v2, and pli0096_v3 based on the earliest non-missing observation on volunteering from the respective survey year.

Across survey years, the response categories differentiate between never, rarely, occasionally, frequently, and daily participation in volunteer activities. These categories have been recoded into a five-point measure, with higher numbers indicating more frequent activities, as follows: never – 0, rarely – 1, occasionally – 2, frequently – 3, and very frequently – 4. Coding items into these categories preserves their ordinal quality while making interpretation easier in the thesis text and figures.

Because volunteering is not observed in every survey year, the moderator is not simply an annual lag. Instead, for each individual with an observed retirement transition, pre-retirement volunteering is measured as the most recent non-missing volunteering observation recorded within the one- to three-year window before retirement. This procedure preserves temporal ordering while accommodating the irregular fielding of the civic engagement module.

The main empirical models use a binary indicator distinguishing respondents with no pre-retirement volunteering from those with any pre-retirement volunteering. A more detailed categorical version of volunteering frequency is used in descriptive analyses and supplementary robustness checks. In the final sample, the volunteering measure is observed predominantly one or two years before retirement, with only a very small share observed three years before retirement. The moderator, therefore, captures volunteering shortly before retirement for almost the entire final sample while preserving temporal ordering relative to the transition.

4.3 Outcome Variable

The dependent variable in this study is self-reported health (SRH), captured by the SOEP variable ple0008. The respondents rate their current general health on a five-point scale from 1 (very good) to 5 (bad). Reverse-coding the variable results in values indicating higher levels of health for respondents: srh_rev = 6 – ple0008. Consequently, a scale from 1 to 5 will be developed, with higher scores indicating better self-assessed health.

Self-assessed health is a broadly defined yet popular measure used in the health sciences and aging studies. In the final analysis sample, the mean value of reversed SRH will be about 3.1, with most respondents being located in the middle of the scale.

For robustness checks, an analysis using the item's original ordinal scale will be conducted via ordered logit regression. The findings are interpreted separately from the linear model estimation.

4.4 Retirement Variables

Retirement is the focal transition that structures the event-time design used in this thesis. It is measured by combining several SOEP indicators that capture current and recent retirement status, including whether the respondent is already retired and whether they report having retired since the beginning of the previous year or during the previous year. In the analysis extract, these indicators are operationalized using plc0311, plb0282_h, plb0282_v1, and pab0005, which are harmonized into a binary retirement-status measure. An individual is coded as retired in a given year if any of these indicators equals 1.

Retirement is treated as an absorbing state. Once an individual is first observed as retired, all subsequent observations are coded as retired. Based on this measure, a retirement-event indicator identifies the first observed transition from non-retired to retired.

For each respondent with an observed transition, the retirement year is defined as the first survey year in which the retirement event occurs. Event time is then defined as the survey year minus the first retirement year, so that retirement year 0 serves as the reference point; negative values indicate pre-retirement years, and positive values indicate post-retirement years. The analysis is restricted to the interval from -2 to +5 years relative to retirement.

4.5 Covariates and Control Variables

In this analysis, the primary covariates used are age, gender, migration background, education, and income. For the empirical analysis, age, gender, migration background, and education are treated as control variables in the pooled regressions, whereas fixed-effects regressions account for all individual-time-invariant variables and thus do not estimate their effects.

Age is defined as the number of years, as the difference between the survey year and the year of birth. Since the analysis considers retirement transitions in later periods of working life, only individuals aged from 50 to 70 were included in the sample. Gender is specified as male or female.

Educational achievement is defined using the generated ISCED-1997 indicator variable pgisced97 from the SOEP pgen data file. For all individuals, the highest reported education category prior to retirement is used as their time-invariant value for educational achievement. In the pooled models, the education variable is included as a control variable representing long-term socioeconomic status. According to the official documentation, the generated indicators for education, such as pgisced97 and pgisced11 from the SOEP pgen data, are considered internationally comparable education indicators; they classify respondents' highest attained degree.

Household income is determined using the harmonized variable hlc0005_h, which represents households' monthly net income. It is also supported by the source variables hlc0005_v1 and hlc0005_v2, wherever required, in the analytical data set. Income is scaled to thousands of euros and centered in the pooled models, but discretion should be exercised since income changes mechanically in response to retirement.

4.6 Sample Construction

The analysis sample is constructed using several stages. First, the SOEP person-year base is filtered for the period 2003 to 2023. Secondly, the respondents are filtered by age to those between 50 and 70. Thirdly, only those who experience a first retirement transition are kept in the data. Fourthly, the person-year base is restricted to event windows from -2 to +5 with respect to retirement. Lastly, the individual level is restricted to individuals who retain a pre-retirement volunteering measure for one to three years before retirement.

This procedure results in a well-fitted sample with respect to the research question on how health changes surrounding retirement, as well as on the differential impact of prior volunteering. This procedure also reveals that the analysis is conditional on having experienced the transition to retirement and on having valid pre-retirement volunteering information. A procedure for sample restrictions is shown in Appendix Table A1.

4.7 Descriptive Profile of the Final Sample

The sample for the analysis is slightly unbalanced, but towards female individuals. The mean age at the descriptive baseline is 60.6 years, and 13% have a migration background. The average net monthly income per household is around 3,076 euros.

Most people have high educational attainment, falling into the highest ISCED categories. Also, the final sample is unbalanced towards non-volunteers, with about 67% non-volunteers and about 33% having volunteered at least once. Thus, there is empirical evidence of a substantive difference between people with no volunteering experience and those who have engaged in volunteering, whereas the frequency of volunteering is a supplementary variable.

The event times still have high person-years even as they approach the post-retirement period within the selected time interval. This shows that using event times with years right before and after retirement is a proper choice.

5. Empirical Strategy

5.1 Overview

The empirical approach examines changes in self-reported health around the transition into retirement and how the pattern of change varies with volunteer activities before retirement. It is important to understand that the approach focuses on describing how health changes around retirement among those who experienced it, rather than estimating the effect of retirement per se. This is relevant since both retirees and volunteers are selected into these categories.

Given the scope of the analysis, the adopted specification is fairly straightforward in its event-time approach. The first specification uses a pooled OLS model with event-time interactions, and the results are presented using the adjusted predicted values. The second specification employs an event-time model with individual fixed effects for testing within-person effect modification. Additional analyses include an ordered logit, stabilization of the inverse probability weighting for sample selection, and specifications with fixed effects for survey year. Analyses of heterogeneity in results by gender and socioeconomic status, stratified by education level, were also conducted. Joint Wald tests were used to test the significance of the volunteering-by-event-time interaction block compared to zero.

5.2 Event-Time Design

Event Time for each observation with a recorded initial transition to retirement is defined as the number of survey years before or after the year of retirement.

It is equal to zero if it is the year of retirement, negative if it is the year before retirement, and positive if it occurs after the transition. The study will use event times only within the range -2 to +5. By doing so, we can focus on health dynamics near the transition and minimize reliance on data far from retirement, since the sample size is smaller in these areas and general age-related health changes are harder to distinguish from the effects of retirement.

The period without a reference in event-time models corresponds to event time –1, being the period just prior to retirement. Therefore, coefficients and predictions must be understood relative to the last observed year prior to retirement, and it makes sense, for interpretation, to consider whether health changes - positively, negatively, or remains the same.

Volunteering prior to retirement is based on the last non-missing volunteering observation made between one and three years prior to retirement. The reason for this decision is that not every year has a volunteering observation, and using an annual lag would artificially reduce the sample size. Moreover, it would not align with fielding the volunteer item within the civic participation module of SOEP. In reality, volunteering observations are made one year prior to retirement for 48.3% of respondents, two years prior for 50.5%, and three years prior for 1.3%.

The interpretation of the coefficients on event-time variables would then involve differences from the year preceding retirement in the sample under analysis, rather than indicating a causal relationship between retiring and health. This is because any gains observed before the transition may be due to anticipation effects or timing issues, among other factors, and are not necessarily attributable to retirement. Therefore, the use of the event-time technique will be instrumental in defining short-term health trends near the transition period and in establishing whether these trends vary with the presence of volunteer work prior to retirement.

5.3 Main Model: Pooled OLS with Event-Time Interactions

The main specification is a pooled OLS regression of reverse-coded self-rated health on event-time indicators, pre-retirement volunteering, and their interaction:

$$SRH_{it} = \alpha + \sum_k \beta_k D_{it}^k + \gamma Vol_i + \sum_k \delta_k (D_{it}^k x\, Vol_i) + X_{it}\theta + \varepsilon_{it}$$

where SRH_{it} denotes reverse-coded self-rated health for individual i in year t, D_{it}^k is an indicator for event time k, Vol_i denotes pre-retirement volunteering, and X_{it} includes covariates such as age, sex, migration background, education, and household income.

In the main analysis, event time is treated as a factor variable and interacted with a binary variable that captures the presence of pre-retirement volunteering for any frequency versus no volunteering at all for the sample of interest. The binary variable is used as the key moderator in these models since volunteering occurs primarily among non-volunteers, whereas extremely frequent volunteering is less common. While a more refined definition of volunteering frequency will be reserved for additional analysis, the binary comparison offers the most compelling distinction to consider.

The proposed model allows the mean level of self-reported health to vary freely with event time before retirement and examines how the shape of the trajectory differs between pre-retirement volunteers and non-volunteers. The substantive interpretation of this specification can be found in the interaction coefficients.

The pooled OLS model is considered the primary model specification for three reasons. Firstly, it is easy to understand and particularly suitable for graphically depicting predicted values. Secondly, it clearly aligns with the research question, namely, whether health pathways before and after retirement vary with pre-retirement participation in volunteering activities, rather than whether volunteering causes better health outcomes. Thirdly, it serves as a clear point of comparison for assessing the fixed effects and robustness models. Covariates are centered, income is measured in thousands of euros, and, since the sample is based on person-years, the standard errors are clustered at the individual level.

5.4 Event-Time Model with Individual Fixed Effects

To strengthen the longitudinal interpretation, the analysis also estimates an event-time model with individual fixed effects:

$$SRH_{it} = \alpha_i + \sum_k \beta_k D_{it}^k + \sum_k \delta_k (D_{it}^k x \, Vol_i) + X_{it}\theta + \varepsilon_{it}$$

where α_i denotes individual fixed effects.

This model captures all time-invariant characteristics of individuals, both observed and unobserved, and thus analyzes changes in self-assessed health within individuals before and after retirement. Since the pre-retirement variable volunteering is constant within the analytic period, it does not vary much and thus gets absorbed into the individual fixed effects. However, the interaction between pre-retirement volunteering and event time remains identifiable as it varies from year to year for an individual.

In fact, this model is very helpful for longitudinal analysis. It can capture systematic changes in health before retirement and differences in post-retirement trajectories due to volunteering.

The fixed-effects model is not presented as a fully causal design. Rather, it serves as a stronger test of whether the patterns observed in the pooled specification persist when the analysis is restricted to within-person change.

5.5 Robustness and Sensitivity Checks

Since self-rated health was measured on a five-point ordinal scale, the interaction model is estimated again using ordered logit as a robustness check of the measure used. The research issue of interest is whether the sign and pattern of the primary conclusions depend on the assumption that the reverse-coded self-rated health variable is continuous.

The second sensitivity analysis involves estimating the interaction model with stabilized inverse-probability weights, where the weights are obtained from a selection model that uses pre-retirement factors to predict membership in the sample used in estimation. While this procedure does not solve the causal identification problem, it offers an assessment of whether the primary conclusions can be driven solely by differences in observability and non-missingness.

In the third robustness test, the primary pooled and fixed-effects binary models are estimated again with survey-year fixed effects. The purpose of this analysis is to determine whether the conclusions reached thus far are robust to any disturbances during the calendar years, as well as more general changes in the period itself that affect all individuals involved.

Last but not least, Wald joint tests are conducted to see whether the complete set of interactions between volunteering and event time is significantly different from zero.

5.6 Gender and Socioeconomic Heterogeneity

Indeed, gender is considered to be a substantively interesting variable in relation to heterogeneity rather than simply another routine control variable. As a matter of fact, there is theoretical ground for this, namely, the assumption that gender differences in employment and caring experiences, as well as the role of work identity among older adults of either sex, lead to distinct meanings of retirement for men and women. On the other hand, there is a sample construction problem as well, since the analysis uses only subjects who have undergone retirement and for whom data on their volunteering activities before retirement are available.

In the main pooled estimates, gender is treated as a control variable, while the descriptive analysis compares sample characteristics, volunteering, and health trends by gender and estimates gender-separate models. In this way, the problem of gender heterogeneity is addressed in this thesis while avoiding unnecessary complexity in the main regression models. Finally, the descriptive retention analysis indicates that the difference in the sample's gender composition was not large.

In this analysis of socioeconomic status stratification, as discussed in the theoretical section, education will be used as a measure of socioeconomic status. The reason for selecting education as an SES variable is its relative stability over the life cycle and its lower vulnerability to change upon entering retirement compared to household income. These additional models should be understood with some reservations about their application; their main objective is to determine whether the relationship under study varies across social groups.

5.7 Interpretation and Limits

This strategy allows for temporal ordering and reduces, though it does not completely rule out, problems of reverse causation. Assessing volunteering prior to retirement is one way to minimize the likelihood of a correlation between health and volunteering in response to life changes associated with retirement. Nevertheless, the analysis remains based on an observational framework.

The inclusion of the within-person change specification addresses some aspects of this problem by isolating the within-person effects and eliminating individual heterogeneity from the equation. Nevertheless, the fixed-effects specification cannot account for time-varying unobservables that affect both outcomes. In this regard, the results obtained are not considered causal estimates of the effect of volunteering on health.

Another limitation is that selection bias in the construction of the analytical sample means that only those who transition to retirement during the observation period and have a valid measure of volunteering prior to retirement will be included in the analysis. Thus, the selected sample is highly appropriate for examining trajectory patterns regarding retirement, but does not necessarily reflect the experiences of all older Germans.

Regarding the research questions, one should be careful not to view the findings presented herein as causal evidence about volunteering and health trajectories, but rather as patterned evidence. This is completely consistent with the research objectives, as the primary question of interest was whether individuals with different pre-retirement civic activities exhibited distinct health patterns around retirement.

6. Results

6.1 Descriptive Statistics and Sample Characteristics

This section first describes the analytical sample and then turns to the estimated health trajectories around retirement. As outlined in the data section, the final analytical panel is restricted to individuals aged 50 to 70 who experience an observed first retirement transition between 2003 and 2023 and who have a valid pre-retirement volunteering measure within the one- to three-year window before retirement. The substantive focus is therefore not on older adults in Germany in general, but on a retirement-event sample aligned with the study's central research question.

For the regression models, the estimation sample is slightly smaller than the full analytical panel because observations with missing covariate values are excluded at the model stage. The main estimation sample contains 8,314 individuals and 46,349 person-years. Mean reverse-coded self-rated health is approximately 3.1, indicating that most respondents are concentrated in the middle of the health distribution rather than at the extremes. About one-third of the estimation sample reports some degree of pre-retirement volunteering, and the sample remains nearly balanced by sex.

The profile of descriptives reveals several characteristics that have already been observed in the construction of the sample. Pre-retirement volunteering, even in its most frequent form, is relatively rare among the respondents included in our sample. The majority of participants reported having done no volunteer work prior to retirement, and those who had done so were fewer in number: rare, occasional, or frequent. The proportion of those who volunteer very frequently is marginal. These observations lend credibility to the choice to use the dichotomous volunteering measure as the primary moderator in the empirical analysis, with the more elaborate frequency measure as the secondary moderator.

The trajectories observed above reveal a clear pattern-individuals involved in pre-retirement volunteering rate their health status higher than the control group at all event times. Nonetheless, there are virtually no differences between these two groups in changes in their health during the transition into retirement. This means that pre-retirement volunteering might influence the average health status of people, yet not the process of aging itself in the period following retirement. This observation will prove critical later since the regression results below corroborate it.

Characteristic	N = 8,314[1]
Age	60.6 (5.6)
Sex	
Male	4,086 (49%)
Female	4,228 (51%)
Migration background	1,117 (13%)
Highest observed ISCED category	
1	81 (1.0%)
2	672 (8.1%)
3	4,402 (53%)
4	371 (4.5%)
5	662 (8.0%)
6	2,103 (25%)
Monthly household income	3,076 (3,113)
Self-rated health (reversed)	3.10 (0.96)
Any pre-retirement volunteering	
No volunteering	5,580 (67%)
Any volunteering	2,734 (33%)
[1] Mean (SD); n (%)	

Table 1. Descriptive characteristics of the final analytic sample

6.2 Main Results: Pooled OLS with Event-Time Interactions

Table A2 and Figure 2 present the results from the pooled OLS estimation, which serve as the benchmark specification in the analysis. As shown, there is a significant positive correlation between past volunteering and SRH. Using the binary specification, individuals who have ever volunteered prior to retirement receive, on average, an extra

0.11-point score on the reverse-scored five-point SRH index compared to those without volunteering, controlling for the included variables. This difference is slight but persistent, given the relative compression of the dependent variable.

The graph of adjusted predicted values is much more revealing than the point estimates on event-time effects. Throughout the event time window, individuals who have previously volunteered have higher SRH predictions than non-volunteers. Meanwhile, the paths are roughly parallel. Pooled interaction effects across the event time window are small and insignificant. This suggests there is no substantial evidence that the volunteers follow a markedly different path of health development in relation to retirement. This observation is also consistent with the joint tests presented in Appendix Table A5.

This distinction is critical for the analysis of the pooled data set. The model predicts that pre-retirement volunteering generally leads to better health, but it fails to find compelling evidence that volunteering mitigates short-run health declines due to retirement.

The frequency model confirms the interpretation of the main effect model without changing it. Respondents who engaged in pre-retirement volunteering on a rare, occasional, or regular basis all showed higher average health than non-volunteers. Nevertheless, interaction terms remain weak and uncertain for frequent volunteers. Because only a few people in the sample engaged in frequent volunteer work, this result should be considered supplementary evidence.

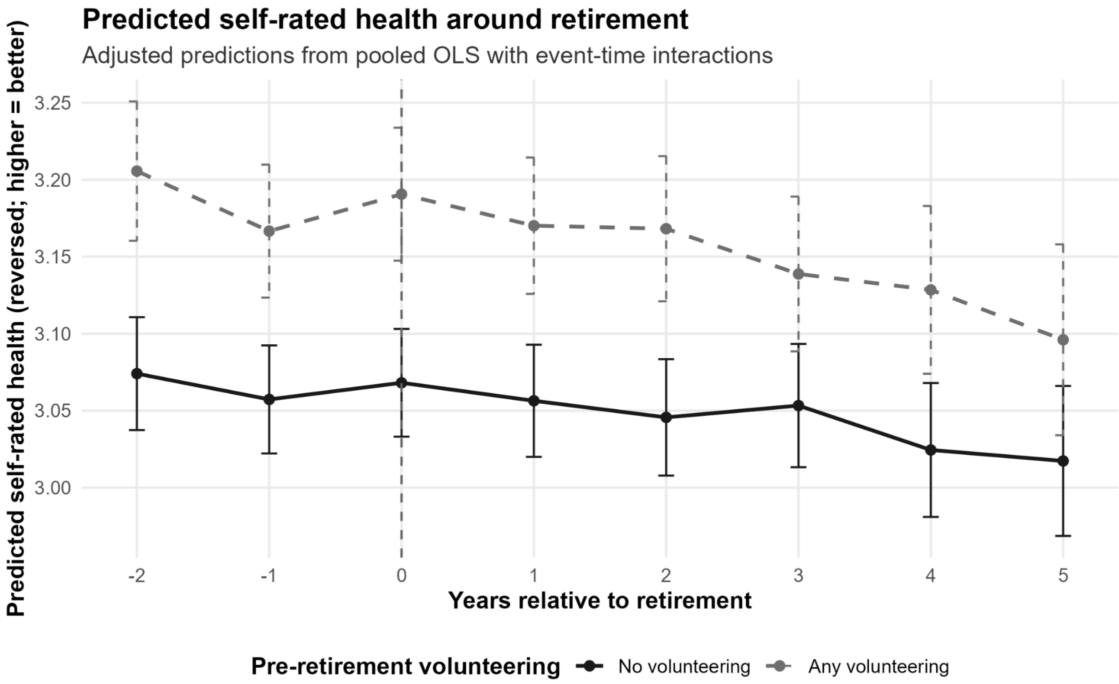


Figure 2. Predicted self-rated health around retirement

6.3 Event-Time Model with Individual Fixed Effects

The fixed-effects event-time model imposes a stricter longitudinal test by stripping away all time-invariant person-specific heterogeneity, leaving only within-person changes relative to the reference year preceding retirement. This is the true test of whether the patterns found in the pooled OLS regression remain valid after eliminating permanent differences between individuals.

The fixed-effects regression results indicate short-term within-person changes in self-reported health status near the time of retirement, rather than a steady post-retirement trend. Compared to the reference period event time -1, self-reported health is more positive at event times -2 and near the transition, then becomes less positive over time, particularly at event times 4 and 5. Since the trends had already changed before the transition, the results should be considered as within-person short-term changes near the retirement period.

Notably, the interaction effects between event time and past volunteer work are small and not statistically significant throughout the event period. In this regard, the fixed-effects plot shows that the within-person paths for both groups are nearly identical when controlling for stable between-person differences. As such, the better health status of past volunteers identified in the pooled sample reflects the average health difference between individuals, rather than an individual response to the retirement process.

This finding clarifies the interpretation of the buffering effect in the theoretical model. Namely, people who engage in volunteer activities prior to retirement have a more favorable health status; however, they respond to retirement similarly to those who do not volunteer.

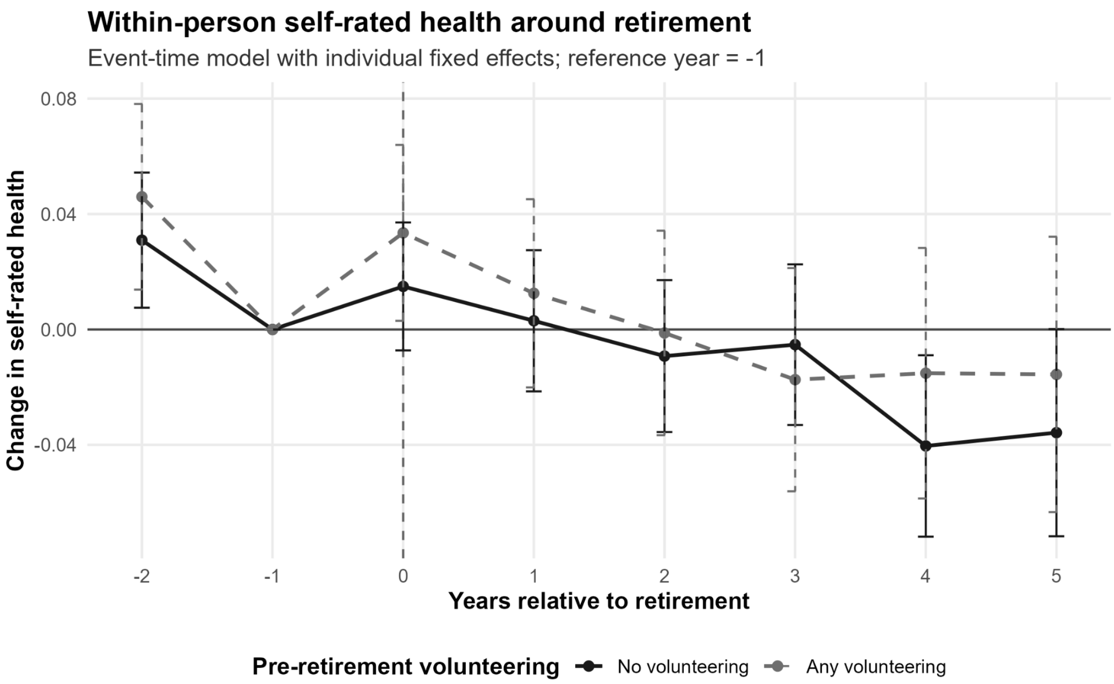


Figure 3. Within-person self-rated health around retirement

6.4 Robustness and Heterogeneity

Robustness tests provide additional justification for the principal conclusion. Firstly, as with the results obtained from the ordered logit model, the same result is replicated by the pooled ordinary least squares regression. In the ordered logit model, prior volunteering remains significant for better SRH, whereas the effect modification term is insignificant. As such, the findings remain robust under the assumption that reverse-coded SRH reflects continuous SRH.

Secondly, an inverse probability weighting robustness test yields nearly identical results and thus serves as a sensitivity test for the sample selection process. Using stabilized inverse-probability weights, the principal finding regarding volunteering remains intact: prior volunteering is significantly associated with better SRH, without introducing any new effects.

Third, the results from models that include survey-year fixed effects are not qualitatively different. There is still a positive effect of past volunteering on self-reported health, although the interaction terms are relatively insignificant. This minimizes the possibility that common shocks or trends in the calendar years play an important role in producing the result of interest.

Fourth, formal tests confirm what is seen in the graphs above. Jointly testing the significance of the interaction block containing volunteering and event time variables shows that this block is not statistically significant in both model specifications considered here, including those that consider only post-retirement interaction terms.

The results of the heterogeneity tests give only limited evidence of marked differences within the subgroups of interest. In the sex-stratified pooled models, previous volunteering is positively associated with self-assessed health status among both men and women; however, the interaction model provides no convincing evidence of substantial differences in the trajectories of health and retirement between the sexes and volunteers. Likewise, while the SES heterogeneity analysis by education reveals a positive effect of higher education on good health, the interaction effect is unconvincing, indicating no significant role of prior volunteering in changes in self-reported health near retirement.

Thus, while providing additional insights into the study results, both robustness and heterogeneity analyses confirm the findings of the previous two models.

6.5 Summary of Findings

Three primary conclusions may be derived from the findings.

First, self-rated health improves slightly during the period surrounding retirement, but deteriorates in the years after retirement. Such trends are especially apparent in the results produced by the fixed-effects approach.

Second, prior volunteering has been found to have a positive effect on self-rated health. It is observed that individuals who engaged in volunteering before retiring evaluate their health more positively than individuals without such experience across all approaches used.

Lastly, and most directly relevant to answering the thesis question, there is little indication that pre-retirement engagement in volunteering significantly buffers health transitions during retirement. Interaction terms are relatively small in magnitude and statistically insignificant, regardless of whether the analysis uses within-individual fixed effects or joint Wald tests. Thus, the findings favor an understanding based on selection or profiling rather than one centered on buffering. Specifically, past volunteers tend to be healthier but experience similar health transitions during the course of their retirement.

From a substantive point of view, this implies that participation in volunteer activities before retirement might serve more to distinguish people who have been healthier and socially advantaged than to facilitate adaptation to retirement.

7. Discussion and Policy Implications

7.1 Interpretation of Findings in Light of Theory

The findings of this thesis suggest that the relationship between volunteering, retirement, and health is more limited and more selective than some of the more optimistic interpretations that the literature imply. Across the descriptive results and pooled models, individuals who reported pre-retirement volunteering exhibited better self-rated health than those who did not volunteer. However, the central finding of the thesis is that there is little evidence that pre-retirement volunteering substantially alters short-run health trajectories after retirement. Taken together, the results point more clearly to a health-level difference between persons than to a trajectory difference within persons across the retirement transition.

While in some respects, this finding aligns with both role and continuity theories, it modifies their implications. Both perspectives argue that the process of retirement may lead to de-structuring of life, loss of purpose and social connectedness and that taking up new roles, such as volunteering, could serve as a way to maintain life continuity and integrity. From this perspective, the higher mean levels of health among those who were engaged in volunteering prior to retirement could be seen as an example of life trajectory characterized by high social involvement, well-established routine, and abundant personal resources. However, according to the findings from the model with fixed effects, once individual differences are controlled for, volunteering does not generate a statistically significant divergence between the health trajectories before and after retirement.

In this regard, the results match up best with the selection side of the literature on volunteering and its health implications. Past research usually found a correlation between volunteering and improved health, wellbeing, and successful aging, but more recent research based on longitudinal data and self-selection has indicated that healthier, better integrated, and richer people tend to become volunteers. In contrast, our results suggest a pattern that can be explained according to the second perspective. Volunteering before retirement seems to be part of a more general phenomenon of social integration, habitual behavior, and accumulated advantage that leads to improved health, but not to any distinct short-term changes in health status upon retirement.

However, the above insights should not be interpreted as an indication that volunteering does not matter at all. Instead, it implies that the role of volunteering in relation to health in this setting can be more about signaling a biography characterized by social involvement and social ties than contributing a particular health benefit at retirement. Thus, the key novelty of the current thesis consists in going beyond the idea that volunteering positively affects health and proving that the empirical association observed for Germany can be best explained in terms of selection, persistence, and cumulation. Again, the findings regarding heterogeneity support the same conclusion. While there might be some descriptive differences in regard to sex and socio-economic status defined in terms of educational attainment, no clear patterns emerge from the interaction models in terms of the systematic influence of volunteering on health development after retirement. In other words, the evidence produced is more consistent with recent panel studies focusing on self-selection than with views on the impact of volunteering on health dynamics.

7.2 Implications for Healthy Aging, Civic Infrastructure, and German Policy

The implications of this study's findings do not suggest that volunteering needs to be considered a separate health intervention strategy. This is because such an implication will be misleading. It can instead be concluded that volunteering needs to be part of a bigger picture that considers healthy aging, participation, and connectedness. Indeed, the present health policy framework in Germany has incorporated the idea of including all these aspects. This is because, according to the Federal Ministry of Health, "healthy aging does not depend just on health care and health promotion but also on social inclusion, participation, local circumstances, and a smooth transition into retirement". The current findings fit well within this policy paradigm because people who were involved before retirement seem to retire in better health (Federal Ministry of Health, 2025).

On the other hand, the thesis offers another important lesson for policymakers. If the relationship between volunteering and health is based primarily on a healthier selection process among volunteer participants, the policy can no longer expect that encouraging participation in such activities would lead to an equal improvement in the well-being of all citizens. It is especially true concerning today’s civic engagement initiative in Germany, where both the engagement strategy at the federal level as well as the civic engagement policy by the ministry involve strengthening, enabling, and recognizing diverse engagement while removing existing structural obstacles to civic engagement, as well as providing sustained support for volunteers and civic infrastructures and fostering recognition (BMBFSFJ, 2025c).

The Fourth Civic Engagement Report makes this statement even clearer. The report explains that civic engagement in Germany is unequal across income, education, employment, and citizenship status, and that participation depends on various thresholds that may not always be visible and include factors such as money, time, discrimination, representation, bureaucratic processes, and location. It further highlights that although civic engagement can lead to cooperation, participation, and positive well-being, people with limited opportunities to engage in these activities face greater obstacles to involvement in civic life (BMBFSFJ, 2025a). These findings have tremendous significance for interpreting the study results. Volunteering is associated with good health in retirement, but because opportunities are limited, it cannot be considered a resource for everyone.

The second connection between this topic and public policy lies within the issue of loneliness and social connectedness. For the federal, loneliness strategy implemented in 2023, loneliness is seen as a cross-generational policy challenge, with the aim of enhancing social connectedness and interaction across generations. In this context, it becomes evident from the monitoring efforts of the BMFSFJ and its publicity campaigns that the policy instrument in question is not limited to a certain type of action or activity; rather, it includes numerous measures, public activities, model projects, and low-threshold activities in addition to programs for older people. As mentioned before, the relevance of the strategy in this case lies in the fact that the federal evidence base on loneliness uses SOEP data, as evidenced by the 2024/2025 barometer literature (BMBFSFJ, 2023). Although loneliness is not tested in this thesis, the results may contribute to the ongoing debate on the significance of civic engagement in the context of social connectedness and healthy aging.

Thus, the best interpretation of the policy implications from this study would have to involve several layers. First, volunteering can be thought of as only one element of a broader policy framework on health and aging and civic engagement, rather than an alternative to health care, financial security, and antipoverty initiatives. Second, since volunteerism is not available on an equal basis, policies promoting the engagement of older adults in civic activities also need to take into account efforts to reduce existing barriers that make volunteering more difficult for some individuals than for others. Third, policies on retirement, health and aging, civic engagement, and loneliness cannot be considered in isolation from one another.

7.3 Policy Recommendations

Three policy recommendations follow from this interpretation.

First, policymakers must acknowledge the period just before and after retirement as an important stage when citizens can engage socially. Local authorities, senior care providers, organizations, and health stakeholders in the community can establish easy-access referral systems for soon-to-be retirees to participate in volunteer work and community events. This approach seems to fit well with the findings discussed above better than the call for volunteerism among retired citizens.

Second, civic engagement policies should consider access equality. As stated in the Fourth Civic Engagement Report, personal finances, free time, discrimination, and organizational culture determine whether citizens have opportunities to participate. It indicates that concrete actions are needed, such as covering transportation expenses, allowing flexible participation periods, adopting hybrid participation approaches, engaging disadvantaged minorities, and implementing practices that eliminate exclusion from civic engagement programs (BMBFSFJ, 2025a).

Third, strategies for promoting healthy aging and reducing loneliness should more actively engage with the idea of civic infrastructure. This is already done through current federal policies that encourage local projects, participatory approaches, and sites of encounter. The significance of this thesis is that civic infrastructure can be seen not only as normatively appealing but also as a potentially useful framework for understanding how individuals may view an important life-stage transition, such as retirement (BMBFSFJ, 2026).

7.4 Limitations and Directions for Future Research

However, there are several limitations in this thesis that should be considered. To begin with, the study is limited to a sample of older individuals who experienced retirement, rather than the entire population of older people. As a result, while the study may accurately identify health trends surrounding the retirement event, its findings cannot be directly extrapolated to all cases of late-life health trends. Additionally, pre-retirement volunteering is assessed based on the most recent observation from the period of one to three years preceding retirement. Although such an approach is justified by the lack of annual observations for the SOEP volunteering question, it is not an alternative.

Then the dependent variable is self-rated health. It is an important variable, often studied across research, but it does not specifically refer to mechanisms like loneliness, depression, social support, and/or functional limitations. Hence, it will be impossible for the thesis to investigate if the mechanism underlying volunteering and health outcomes is the same for those listed above or a combination of any/all of the above. Lastly, although the method of investigation in this paper enhances temporal logic and employs a fixed-effects model and robustness checks, this thesis remains based on an observational approach. It is quite possible that healthier people and those with more advantages volunteer before retiring.

There are several clear avenues for future research. The first would involve taking a closer look at measures associated with the suggested mechanisms. This should be especially useful in light of the federal German loneliness strategy and loneliness barometer based on the SOEP (BMBFSFJ, 2025b). The second would involve studying the different types of post-retirement volunteering trajectories, namely continuation, entry into volunteer activity after retirement, exit from volunteering, and intensification of volunteer activity after retirement. Third, one could incorporate event-time designs with causal methods or evaluate actual civic participation programs to determine when such programs make a difference in improving older adults' quality of life. Finally, further work needs to focus on unequal access. If volunteering is indeed unequally distributed among older people, then knowing who can actually benefit from civic participation is equally as important as finding average associations.

8. Conclusion

The thesis analyzes the evolution of self-rated health before and after the transition into retirement in Germany, as well as if health trajectories differ depending on pre-retirement volunteering. Based on SOEP panel data under the assumption of event-time specification, it was demonstrated that there exists a strong descriptive association between pre-retirement volunteering and self-reported health in that volunteers exhibit higher levels of self-rated health during the process of retirement. However, the empirical evidence for significant changes in the health trajectories following retirement induced by volunteering remains rather weak, and a health level difference is indicated instead.

Thus, the significance of the present thesis lies in its empiricism and interpretation. On the one hand, it synthesizes two streams of literature which typically are analyzed separately; namely, the health transition at retirement and the link between volunteering and quality of life after retirement. The results indicate that in the case of Germany, rather than being a buffer against changes in health status, volunteering seems to play a role in life-course processes of selection, prior social integration, and cumulated advantages. Such an interpretation is compatible with recently published evidence based on panel data which finds that healthier people who enjoy greater social integration tend to volunteer more frequently. However, at the same time, the findings clearly point out that volunteering should still be considered a vital element in the social fabric for a healthy older age in Germany despite not being a significant protective factor.

With respect to policy recommendations, the findings can be considered to have an important implication, albeit a careful one. If the link between volunteering and well-being rests largely in the fact that volunteering is an integral part of biographies of social engagement, then policies should concentrate on the creation of conditions which facilitate such engagement in social contexts. In the case of Germany, which already has a rich civic associational life, this would mean the elimination of social obstacles, increased possibilities for civic engagement locally, and the understanding that volunteering is not an equally accessible option for everybody. Therefore, the significance of volunteering is not necessarily its potential impact on shaping retirement patterns, but the fact that it is a part of a larger social and civic world within which people thrive and live more fulfilling lives.

Bibliography

Adler, R. P., & Goggin, J. (2005). What Do We Mean By “Civic Engagement”? *Journal of Transformative Education*, *3*(3), 236–253. https://doi.org/10.1177/1541344605276792

Alscher, M. (2024, November 6). *Zivilgesellschaftliche -Organisationen als Infrastruktur des Zivilengagements | Sozialbericht 2024*. bpb.de. https://www.bpb.de/kurz-knapp/zahlen-und-fakten/sozialbericht-2024/553372/zivilgesellschaftliche-organisationen-als-infrastruktur-des-zivilengagements/

American Psychological Association. (2006). *Civic Engagement*. Https://Www.Apa.Org. https://www.apa.org/education-career/undergrad/civic-engagement

Anderson, N. D., Damianakis, T., Kröger, E., Wagner, L. M., Dawson, D. R., Binns, M. A., Bernstein, S., Caspi, E., & Cook, S. L. (2014). The benefits associated with volunteering among seniors: A critical review and recommendations for future research. *Psychological Bulletin*, *140*(6), 1505–1533. https://doi.org/10.1037/a0037610

Ballard, P. J., Hoyt, L. T., & Pachucki, M. C. (2019). Impacts of Adolescent and Young Adult Civic Engagement on Health and Socioeconomic Status in Adulthood. *Child Development*, *90*(4), 1138–1154. https://doi.org/10.1111/cdev.12998

Bloom, D., Canning, D., & Moore, M. (2007). *A Theory of Retirement* (No. W13630; p. w13630). National Bureau of Economic Research. https://doi.org/10.3386/w13630

BMBFSFJ. (2025a). *Fourth Civic Engagement Report*. https://www.bmbfsfj.bund.de/bmbfsfj/meta/en/publications-en/fourth-civic-engagement-report-key-findings-264412

BMBFSFJ. (2025b, December 4). *Civic Engagement Policy*. BMBFSFJ. https://www.bmbfsfj.bund.de/bmbfsfj/meta/en/civic-engagement-policy-112114

BMBFSFJ. (2026, March 9). *Modellprojekte gegen Einsamkeit*. https://www.bmbfsfj.bund.de/bmbfsfj/themen/engagement-und-gesellschaft/allianz-gegen-einsamkeit/modellprojekte-gegen-einsamkeit-228616

Bogaard, L. V. D., Henkens, K., & Kalmijn, M. (2014). So now what? Effects of retirement on civic engagement. *Ageing & Society*, *34*(7), 1170–1192. https://doi.org/10.1017/S0144686X13000019

Brown, A. G. M., Hudson, L. B., Chui, K., Metayer, N., Lebron-Torres, N., Seguin, R. A., & Folta, S. C. (2017). Improving heart health among Black/African American women using civic engagement: A pilot study. *BMC Public Health*, *17*(1), 112. https://doi.org/10.1186/s12889-016-3964-2

Cacioppo, J. T., & Cacioppo, S. (2014). Social Relationships and Health: The Toxic Effects of Perceived Social Isolation. *Social and Personality Psychology Compass*, *8*(2), 58–72. https://doi.org/10.1111/spc3.12087

Cohen, S., & Wills, T. A. (1985). Stress, social support, and the buffering hypothesis. *Psychological Bulletin*, *98*(2), 310–357. https://doi.org/10.1037/0033-2909.98.2.310

Damon, W. (2004). What is Positive Youth Development? *The ANNALS of the American Academy of Political and Social Science*, *591*(1), 13–24. https://doi.org/10.1177/0002716203260092

de Wit, A., Qu, H., & Bekkers, R. (2022). The health advantage of volunteering is larger for older and less healthy volunteers in Europe: A mega-analysis. *European Journal of Ageing*, *19*(4), 1189–1200. https://doi.org/10.1007/s10433-022-00691-5

DIW. (2026). *Socio-Economic Panel (SOEP), data from 1984-2024 (SOEP-Core v41, EU Edition).* [Text]. DIW Berlin. https://doi.org/10.5684/soep.core.v41eu

Eibich, P. (2015). Understanding the effect of retirement on health: Mechanisms and heterogeneity. *Journal of Health Economics*, *43*, 1–12. https://doi.org/10.1016/j.jhealeco.2015.05.001

Eibich, P., Lorenti, A., & Mosca, I. (2022). Does retirement affect voluntary work provision? Evidence from Europe and the U.S. *Labour Economics*, *76*, 102185. https://doi.org/10.1016/j.labeco.2022.102185

Erlinghagen, M. (2010). Volunteering After Retirement: Evidence from German panel data. *European Societies*, *12*(5), 603–625. https://doi.org/10.1080/14616691003716902

Evers, A. (2019). Diversity and Coherence: Historical Layers of Current Civic Engagement in Germany. *VOLUNTAS: International Journal of Voluntary and Nonprofit Organizations*, *30*(1), 41–53. https://doi.org/10.1007/s11266-018-00066-4

Filges, T., Siren, A., Fridberg, T., & Nielsen, B. C. V. (2020). Voluntary work for the physical and mental health of older volunteers: A systematic review. *Campbell Systematic Reviews*, *16*(4), cl2.1124. https://doi.org/10.1002/cl2.1124

Fried, L. P., Carlson, M. C., Freedman, M., Frick, K. D., Glass, T. A., Hill, J., McGill, S., Rebok, G. W., Seeman, T., Tielsch, J., Wasik, B. A., & Zeger, S. (2004). A social model for health promotion for an aging population: Initial evidence on the experience corps model. *Journal of Urban Health*, *81*(1), 64–78. https://doi.org/10.1093/jurban/jth094

Goebel, J., Grabka, M. M., Liebig, S., Kroh, M., Richter, D., Schröder, C., & Schupp, J. (2019). The German Socio-Economic Panel (SOEP). *Jahrbücher Für Nationalökonomie Und Statistik*, *239*(2), 345–360. https://doi.org/10.1515/jbnst-2018-0022

Greenfield, E. A., & Marks, N. F. (2004). Formal volunteering as a protective factor for older adults' psychological well-being. *The Journals of Gerontology. Series B, Psychological Sciences and Social Sciences*, *59*(5), S258-264. https://doi.org/10.1093/geronb/59.5.s258

Hämäläinen, H., Tanskanen, A. O., Arpino, B., Solé-Auró, A., & Danielsbacka, M. (2024). Is Transition to Retirement Associated With Volunteering?

Longitudinal Evidence from Europe. *Research on Aging*, *46*(9–10), 509–520. https://doi.org/10.1177/01640275241251786

Han, S. H., Kim, K., & Burr, J. A. (2018). Stress-buffering effects of volunteering on salivary cortisol: Results from a daily diary study. *Social Science & Medicine*, *201*, 120–126. https://doi.org/10.1016/j.socscimed.2018.02.011

Hanemann, F. (2017). *Working Conditions, Retirement and Health – Longitudinal Evidence from Europe and the US* (SSRN Scholarly Paper No. 3060413). Social Science Research Network. https://doi.org/10.2139/ssrn.3060413

Hansen, T., Aartsen, M., Slagsvold, B., & Deindl, C. (2018). Dynamics of Volunteering and Life Satisfaction in Midlife and Old Age: Findings from 12 European Countries. *Social Sciences*, *7*(5). https://doi.org/10.3390/socsci7050078

Harris, A. H. S., & Thoresen, C. E. (2005). Volunteering is Associated with Delayed Mortality in Older People: Analysis of the Longitudinal Study of Aging. *Journal of Health Psychology*, *10*(6), 739–752. https://doi.org/10.1177/1359105305057310

Hayes, L. G., Stangl, M., Guo, M., & Wang, Y. (2025). Considering Community: Examining Civic Engagement and Well-Being in Later Life. *Innovation in Aging*, *9*(Supplement_2), igaf122.2779. https://doi.org/10.1093/geroni/igaf122.2779

Heaven, B., Brown, L. J. e., White, M., Errington, L., Mathers, J. C., & Moffatt, S. (2013). Supporting Well-Being in Retirement through Meaningful Social

Roles: Systematic Review of Intervention Studies. *The Milbank Quarterly*, *91*(2), 222–287. https://doi.org/10.1111/milq.12013

Heller-Sahlgren, G. (2012). *Work 'Til You Drop: Short- and Longer-Term Health Effects of Retirement in Europe* (SSRN Scholarly Paper No. 2153191). Social Science Research Network. https://doi.org/10.2139/ssrn.2153191

Hershberg, R. M., Johnson, S. K., DeSouza, L. M., Hunter, C. J., & Zaff, J. (2015). Promoting Contribution Among Youth: Implications from Positive Youth Development Research for Youth Development Programs. In E. P. Bowers, G. J. Geldhof, S. K. Johnson, L. J. Hilliard, R. M. Hershberg, J. V. Lerner, & R. M. Lerner (Eds.), *Promoting Positive Youth Development: Lessons from the 4-H Study* (pp. 211–228). Springer International Publishing. https://doi.org/10.1007/978-3-319-17166-1_11

Hobfoll, S. E. (2011). Conservation of resources theory: Its implication for stress, health, and resilience. In *The Oxford handbook of stress, health, and coping* (pp. 127–147). Oxford University Press.

Jenkinson, C. E., Dickens, A. P., Jones, K., Thompson-Coon, J., Taylor, R. S., Rogers, M., Bambra, C. L., Lang, I., & Richards, S. H. (2013). Is volunteering a public health intervention? A systematic review and meta-analysis of the health and survival of volunteers. *BMC Public Health*, *13*, 773. https://doi.org/10.1186/1471-2458-13-773

Kaskie, B., Imhof, S., Cavanaugh, J., & Culp, K. (2008). Civic Engagement as a Retirement Role for Aging Americans. *The Gerontologist*, *48*(3), 368–377. https://doi.org/10.1093/geront/48.3.368

Keyes, C. L. M., & Ryff, C. D. (1998). Generativity in adult lives: Social structural contours and quality of life consequences. In *Generativity and adult development: How and why we care for the next generation* (pp. 227–263). American Psychological Association. https://doi.org/10.1037/10288-007

Kim, E. S., & Konrath, S. H. (2016). Volunteering is prospectively associated with health care use among older adults. *Social Science & Medicine*, *149*, 122–129. https://doi.org/10.1016/j.socscimed.2015.11.043

Kim, E. S., Whillans, A. V., Lee, M. T., Chen, Y., & VanderWeele, T. J. (2020). Volunteering and Subsequent Health and Well-Being in Older Adults: An Outcome-Wide Longitudinal Approach. *American Journal of Preventive Medicine*, *59*(2), 176–186. https://doi.org/10.1016/j.amepre.2020.03.004

Kim, S., Halvorsen, C., & Han, S. H. (2023). Volunteering and Changes in Cardiovascular Biomarkers: Longitudinal Evidence From the Health and Retirement Study. *Innovation in Aging*, *7*(5), igad048. https://doi.org/10.1093/geroni/igad048

Kim, S., Halvorsen, C., Potter, C., & Faul, J. (2025). Does volunteering reduce epigenetic age acceleration among retired and working older adults? Results from the Health and Retirement Study. *Social Science & Medicine*, *364*, 117501. https://doi.org/10.1016/j.socscimed.2024.117501

Lehane, G., & Scarlett, S. (2024). Psychological Health Outcomes of Older Adult Volunteers: Findings from The Irish Longitudinal Study on Ageing. *Age and Ageing*, *53*(Supplement_4), afae178.025. https://doi.org/10.1093/ageing/afae178.025

Lerner, R., Lerner, J., Almerigi, J., Theokas, C., Phelps, E., Gestsdottir, S., Naudeau, S., Jelicic, H., Alberts, A., Ma, L., Smith, L., & Bobek, D. (2005). Positive Youth Development, Participation in Community Youth Development Programs, and Community Contributions of Fifth-Grade AdolescentsFindings From the First Wave Of the 4-H Study of Positive Youth Development. *Journal of Early Adolescence - J EARLY ADOLESCENCE*, *25*, 17–71. https://doi.org/10.1177/0272431604272461

Li, Y., & Ferraro, K. F. (2005). Volunteering and depression in later life: Social benefit or selection processes? *Journal of Health and Social Behavior*, *46*(1), 68–84. https://doi.org/10.1177/002214650504600106

Li, Y., & Ferraro, K. F. (2006). Volunteering in Middle and Later Life: Is Health a Benefit, Barrier or Both? *Social Forces*, *85*(1), 497–519. https://doi.org/10.1353/sof.2006.0132

Lühr, M. (2025). Effects of Civic Engagement on Mental Health in Different Age Groups. Insights from Panel Data. In A. Teti, S. Neuderth, M. K. Pavlova, & G. Ziese (Eds.), *Soziale und gesundheitliche Ungleichheit im Alter: Ein Thema, vielfältige Perspektiven* (pp. 125–143). Springer Fachmedien. https://doi.org/10.1007/978-3-658-48005-9_6

Lühr, M., Pavlova, M. K., & Luhmann, M. (2022a). Nonpolitical Versus Political Participation: Longitudinal Associations with Mental Health and Social Well-Being in Different Age Groups. *Social Indicators Research*, *159*(3), 865–884. https://doi.org/10.1007/s11205-021-02777-6

Lühr, M., Pavlova, M. K., & Luhmann, M. (2022b). They are Doing Well, but is it by Doing Good? Pathways from Nonpolitical and Political Volunteering to Subjective Well-Being in Age Comparison. *Journal of Happiness Studies*, *23*(5), 1969–1989. https://doi.org/10.1007/s10902-021-00480-4

Lum, T. Y., & Lightfoot, E. (2005). The Effects of Volunteering on the Physical and Mental Health of Older People. *Research on Aging*, *27*(1), 31–55. https://doi.org/10.1177/0164027504271349

Marmot, M., Friel, S., Bell, R., Houweling, T. A., & Taylor, S. (2008). Closing the gap in a generation: Health equity through action on the social determinants of health. *The Lancet*, *372*(9650), 1661–1669. https://doi.org/10.1016/S0140-6736(08)61690-6

Meier, S., & Stutzer, A. (2008). Is Volunteering Rewarding in Itself? *Economica*, *75*(297), 39–59. https://doi.org/10.1111/j.1468-0335.2007.00597.x

Moen, P. (2001). The gendered life course. In *Handbook of aging and the social sciences, 5th ed* (pp. 179–196). Academic Press.

Moen, P., & Flood, S. (2013). Limited Engagements? Women's and Men's Work/Volunteer Time in the Encore Life Course Stage. *Social Problems*, *60*(2), 206–233. https://doi.org/10.1525/sp.2013.60.2.206

Morrow-Howell, N., Hinterlong, J., Rozario, P. A., & Tang, F. (2003). Effects of Volunteering on the Well-Being of Older Adults. *The Journals of Gerontology: Series B*, *58*(3), S137–S145. https://doi.org/10.1093/geronb/58.3.S137

Nakamura, J. S., Shiba, K., Shi, B., Leong, R. S., VanderWeele, T. J., & Kim, E. S. (2025). How is volunteering associated with reduced mortality? A mediator-wide approach. *Health Psychology*, *44*(5), 518–527. https://doi.org/10.1037/hea0001429

Nascimento, M. de M., Marques, A., Gouveia, É. R., & Ihle, A. (2025). Role of Income, Education, and Physical Activity in Mediating the Relationship Between Self-Rated Health and Political Participation of European Citizens. *Societies*, *15*(10), 281. https://doi.org/10.3390/soc15100281

Nelson, C. D., Sloan, J., & Chandra, A. (2019). *Examining Civic Engagement Links to Health: Findings from the Literature and Implications for a Culture of Health*. https://www.rand.org/pubs/research_reports/RR3163.html

Okun, M. A., Yeung, E. W., & Brown, S. (2013). Volunteering by older adults and risk of mortality: A meta-analysis. *Psychology and Aging*, *28*(2), 564–577. https://doi.org/10.1037/a0031519

Papa, R., Cutuli, G., Principi, A., & Scherer, S. (2019). Health and Volunteering in Europe: A Longitudinal Study. *Research on Aging*, *41*(7), 670–696. https://doi.org/10.1177/0164027519834939

Pavlova, M. K., & Lühr, M. (2023). Volunteering and political participation are differentially associated with eudaimonic and social well-being across age groups and European countries. *PLOS ONE*, *18*(2), e0281354. https://doi.org/10.1371/journal.pone.0281354

Pavlova, M. K., & Silbereisen, R. K. (2012). Participation in voluntary organizations and volunteer work as a compensation for the absence of work or partnership? Evidence from two German samples of younger and older adults. *The Journals of Gerontology. Series B, Psychological Sciences and Social Sciences*, *67*(4), 514–524. https://doi.org/10.1093/geronb/gbs051

Piliavin, J. A., & Siegl, E. (2007). Health Benefits of Volunteering in the Wisconsin Longitudinal Study∗. *Journal of Health and Social Behavior*, *48*(4), 450–464. https://doi.org/10.1177/002214650704800408

Piumatti, G., Magistro, D., Zecca, M., & Esliger, D. W. (2018). The mediation effect of political interest on the connection between social trust and wellbeing among older adults. *Ageing & Society*, *38*(11), 2376–2395. https://doi.org/10.1017/S0144686X1700071X

Putnam, R. D. (2000). Bowling Alone: America's Declining Social Capital. In L. Crothers & C. Lockhart (Eds.), *Culture and Politics: A Reader* (pp. 223–234). Palgrave Macmillan US. https://doi.org/10.1007/978-1-349-62397-6_12

Quick, H. E., & Moen, P. (1998). Gender, employment and retirement quality: A life course approach to the differential experiences of men and women. *Journal of Occupational Health Psychology*, *3*(1), 44–64. https://doi.org/10.1037/1076-8998.3.1.44

Rose, L. (2020). Retirement and health: Evidence from England. *Journal of Health Economics*, *73*, 102352. https://doi.org/10.1016/j.jhealeco.2020.102352

Russell, A. R., Boen, C. E., & Handy, F. (2023). Volunteer Retirement and Well-Being in Older Adults: A Longitudinal Study. *Nonprofit and Voluntary Sector Quarterly*, *52*(5), 1357–1385. https://doi.org/10.1177/08997640221125805

Schreier, H. M. C., Schonert-Reichl, K. A., & Chen, E. (2013). Effect of Volunteering on Risk Factors for Cardiovascular Disease in Adolescents: A Randomized Controlled Trial. *JAMA Pediatrics*, *167*(4), 327–332. https://doi.org/10.1001/jamapediatrics.2013.1100

Schubert, P., Tahmaz, B., & Krimmer, H. (2022). *Vereine in Deutschland im Jahr 2022*.

Serrat, R., Villar, F., Giuliani, M.-F., & Zacarés, J.-J. (2017). Older people's participation in political organizations: The role of generativity and its impact on well-being. *Educational Gerontology*, *43*(3), 128–138. https://doi.org/10.1080/03601277.2016.1269541

Sneed, R. S., & Cohen, S. (2013). A prospective study of volunteerism and hypertension risk in older adults. *Psychology and Aging*, *28*(2), 578–586. https://doi.org/10.1037/a0032718

Taniguchi, H. (2006). Men's and Women's Volunteering: Gender Differences in the Effects of Employment and Family Characteristics. *Nonprofit and Voluntary Sector Quarterly*, *35*(1), 83–101. https://doi.org/10.1177/0899764005282481

Thoits, P. A. (2012). *Role-Identity Salience, Purpose and Meaning in Life, and Well-Being among Volunteers*. https://journals.sagepub.com/doi/10.1177/0190272512459662

Thoits, P. A., & Hewitt, L. N. (2001). Volunteer Work and Well-Being. *Journal of Health and Social Behavior*, *42*(2), 115–131. https://doi.org/10.2307/3090173

Van Ingen, E., & Wilson, J. (2017). I Volunteer, Therefore I am? Factors Affecting Volunteer Role Identity. *Nonprofit and Voluntary Sector Quarterly*, *46*(1), 29–46. https://doi.org/10.1177/0899764016659765

Van Willigen, M. (2000). Differential Benefits of Volunteering Across the Life Course. *The Journals of Gerontology: Series B*, *55*(5), S308–S318. https://doi.org/10.1093/geronb/55.5.S308

Vega-Tinoco, A., Gil-Lacruz, A. I., & Gil-Lacruz, M. (2022). Civic Participation as a Promoter of Well-Being: Comparative Analysis among European Countries. *Social Indicators Research*, *164*(1), 217–237. https://doi.org/10.1007/s11205-022-02947-0

Watts, R. J., Diemer, M. A., & Voight, A. M. (2011). Critical consciousness: Current status and future directions. *New Directions for Child and Adolescent Development*, *2011*(134), 43–57. https://doi.org/10.1002/cd.310

Watts, R. J., Griffith, D. M., & Abdul-Adil, J. (1999). Sociopolitical development as an antidote for oppression—Theory and action. *American Journal of Community Psychology*, *27*(2), 255–271. https://doi.org/10.1023/A:1022839818873

Weziak-Bialowolska, D., Skiba, R., & Bialowolski, P. (2024). Longitudinal reciprocal associations between volunteering, health and well-being: Evidence for middle-aged and older adults in Europe. *The European Journal of Public Health*, *34*(3), 473–481. https://doi.org/10.1093/eurpub/ckae014

Appendix

Appendix Table A0. Main variables, SOEP source items, coding, and analytical role

Construct	SOEP item(s)	Coding / operationalization	Role in analysis	Notes
Self-rated health	ple0008	Original scale 1 = very good to 5 = bad; reverse-coded as srh_rev = 6 - ple0008 so higher values indicate better health	Outcome	Estimated mainly in linear form; original ordinal version used in ordered-logit robustness checks
Pre-retirement volunteering	pli0096_h, pli0096_v1, pli0096_v2, pli0096_v3	First non-missing volunteering item in a survey year; recoded to 0 = never, 1 = rare, 2 = occasional, 3 = frequent, 4 = very frequent	Key explanatory moderator	Most recent non-missing value observed within 1–3 years before retirement
Binary volunteering	Derived from volunteering items	0 = no pre-retirement volunteering; 1 = any pre-retirement volunteering	Main moderator in pooled and FE models	Main contrast used in headline results
Retirement status	plc0311, plb0282_h, plb0282_v1, pab0005	Binary retirement indicator constructed from current and prior retirement reports	Event-time structure	Treated as absorbing after the first retirement observation
Retirement event	Derived from retirement status	First observed transition from non-retired to retired	Event-time structure	Defines retirement year
Event time	Derived	event_time = survey year − first retirement year	Time structure	Restricted to -2 to +5
Age	Derived from survey year and birth year	Age in years	Control in pooled models	Sample restricted to ages 50–70

Sex	SOEP sex variable in analysis extract	Male/female	Control in pooled models	Time-invariant; absorbed in FE models
Migration background	SOEP migration-background variable in analysis extract	Binary indicator	Control in pooled models	Time-invariant; absorbed in FE models
Education	pgisced97	The highest observed ISCED-1997 category observed up to retirement	Control in pooled models	From pgen; treated as time-invariant in pooled models and absorbed in FE models
Household income	hlc0005_h, related to hlc0005_v1, hlc0005_v2	Monthly net household income; entered in pooled models as centered thousands of euros	Control in pooled models	Treated cautiously because retirement may affect income composition
Survey weight	phrf	Person-level cross-sectional / population weight used in descriptive weighting and weighted robustness checks	Weighting/robustness	Not part of the main causal interpretation

Appendix Table A1: Sample restriction flow

Table A1. Sample restriction flow	
SOEP 2003-2023; final analytic window spans 2 years before to 5 years after retirement	
Step	**N_individuals**
Initial SOEP sample (2003-2023)	176,522
Age restriction: 50-70	41,584
Observed retirement transition (first event)	9,296
Event-time window: -2 to +5 years	9,296
Valid pre-retirement volunteering measure	8,314
Final analytical sample	8,314

Appendix Table A2: Full pooled OLS models

Pooled OLS Models with Event-Time Interactions		
	Pooled OLS: Binary volunteering	Pooled OLS: Volunteering frequency
(Intercept)	2.823*** (0.091)	2.822*** (0.091)
event_time_f-2	0.017 (0.013)	0.017 (0.013)
event_time_f0	0.011 (0.012)	0.011 (0.012)
event_time_f1	-0.001 (0.014)	-0.001 (0.014)
event_time_f2	-0.012 (0.015)	-0.011 (0.015)
event_time_f3	-0.004 (0.017)	-0.004 (0.017)

event_time_f4	-0.033+ (0.020)	-0.032+ (0.020)
event_time_f5	-0.040+ (0.023)	-0.039+ (0.023)
vol_pre_binaryAny volunteering	0.109*** (0.023)	
age_c	0.006*** (0.002)	0.006*** (0.002)
sexFemale	0.029+ (0.018)	0.030+ (0.018)
migbackYes	-0.051+ (0.027)	-0.050+ (0.027)
isced_clean_f2	0.013 (0.095)	0.013 (0.095)
isced_clean_f3	0.205* (0.091)	0.205* (0.091)
isced_clean_f4	0.341*** (0.100)	0.341*** (0.100)
isced_clean_f5	0.235* (0.096)	0.235* (0.096)
isced_clean_f6	0.381*** (0.093)	0.381*** (0.093)
isced_clean_fMissing	0.122 (0.210)	0.121 (0.211)
hinc_1k_c	0.035*** (0.008)	0.035*** (0.008)
event_time_f-2 × vol_pre_binaryAny volunteering	0.022 (0.022)	
event_time_f0 × vol_pre_binaryAny volunteering	0.013 (0.020)	
event_time_f1 × vol_pre_binaryAny volunteering	0.004 (0.022)	
event_time_f2 × vol_pre_binaryAny volunteering	0.013 (0.025)	

event_time_f3 × vol_pre_binaryAny volunteering	-0.024 (0.027)	
event_time_f4 × vol_pre_binaryAny volunteering	-0.005 (0.031)	
event_time_f5 × vol_pre_binaryAny volunteering	-0.031 (0.035)	
vol_pre_catRare		0.084* (0.034)
vol_pre_catOccasional		0.105** (0.035)
vol_pre_catFrequent		0.135*** (0.033)
vol_pre_catVery frequent		0.193 (0.131)
event_time_f-2 × vol_pre_catRare		-0.007 (0.031)
event_time_f0 × vol_pre_catRare		0.014 (0.029)
event_time_f1 × vol_pre_catRare		0.011 (0.032)
event_time_f2 × vol_pre_catRare		-0.045 (0.036)
event_time_f3 × vol_pre_catRare		-0.060 (0.040)
event_time_f4 × vol_pre_catRare		-0.018 (0.044)
event_time_f5 × vol_pre_catRare		-0.025 (0.051)
event_time_f-2 × vol_pre_catOccasional		0.037 (0.035)
event_time_f0 × vol_pre_catOccasional		0.025 (0.032)
event_time_f1 × vol_pre_catOccasional		0.051 (0.036)

event_time_f2 × vol_pre_catOccasional		0.089* (0.040)
event_time_f3 × vol_pre_catOccasional		0.035 (0.044)
event_time_f4 × vol_pre_catOccasional		0.039 (0.051)
event_time_f5 × vol_pre_catOccasional		-0.055 (0.059)
event_time_f-2 × vol_pre_catFrequent		0.059+ (0.033)
event_time_f0 × vol_pre_catFrequent		0.019 (0.030)
event_time_f1 × vol_pre_catFrequent		-0.034 (0.033)
event_time_f2 × vol_pre_catFrequent		0.018 (0.037)
event_time_f3 × vol_pre_catFrequent		-0.039 (0.041)
event_time_f4 × vol_pre_catFrequent		-0.028 (0.045)
event_time_f5 × vol_pre_catFrequent		-0.004 (0.052)
event_time_f-2 × vol_pre_catVery frequent		-0.411* (0.178)
event_time_f0 × vol_pre_catVery frequent		-0.420* (0.200)
event_time_f1 × vol_pre_catVery frequent		-0.227 (0.164)
event_time_f2 × vol_pre_catVery frequent		-0.118 (0.162)
event_time_f3 × vol_pre_catVery frequent		0.144 (0.165)
event_time_f4 × vol_pre_catVery frequent		0.058 (0.191)

event_time_f5 × vol_pre_catVery frequent		-0.245 (0.209)
Num.Obs.	46349	46349
R2	0.036	0.037
Standard errors clustered at the individual level. Outcome: reverse-coded self-rated health. Continuous covariates centered; household income entered in thousands.		

Appendix Table A3: Full fixed-effects event-time models

Individual Fixed-Effects Event-Time Models		
	FE event-time: Binary volunteering	**FE event-time: Volunteering frequency**
event_time = -2	0.031** (0.012)	-0.348* (0.175)
event_time = 0	0.015 (0.011)	-0.223 (0.149)
event_time = 1	0.003 (0.012)	-0.199 (0.162)
event_time = 2	-0.009 (0.013)	-0.131 (0.150)
event_time = 3	-0.005 (0.014)	0.099 (0.125)
event_time = 4	-0.040* (0.016)	0.015 (0.143)
event_time = 5	-0.036+ (0.018)	-0.293+ (0.173)
event_time = -2 × vol_any	0.015 (0.020)	
event_time = 0 × vol_any	0.019 (0.019)	
event_time = 1 × vol_any	0.010 (0.021)	

event_time = 2 × vol_any	0.008 (0.023)	
event_time = 3 × vol_any	-0.012 (0.024)	
event_time = 4 × vol_any	0.025 (0.027)	
event_time = 5 × vol_any	0.020 (0.030)	
event_time = -2 × vol_pre_cat = Never		0.379* (0.176)
event_time = -2 × vol_pre_cat = Rare		0.367* (0.177)
event_time = -2 × vol_pre_cat = Occasional		0.409* (0.178)
event_time = -2 × vol_pre_cat = Frequent		0.424* (0.177)
event_time = 0 × vol_pre_cat = Never		0.238 (0.150)
event_time = 0 × vol_pre_cat = Rare		0.261+ (0.152)
event_time = 0 × vol_pre_cat = Occasional		0.272+ (0.152)
event_time = 0 × vol_pre_cat = Frequent		0.249 (0.152)
event_time = 1 × vol_pre_cat = Never		0.202 (0.163)
event_time = 1 × vol_pre_cat = Rare		0.210 (0.165)
event_time = 1 × vol_pre_cat = Occasional		0.272 (0.165)
event_time = 1 × vol_pre_cat = Frequent		0.168 (0.165)
event_time = 2 × vol_pre_cat = Never		0.122 (0.151)

event_time = 2 × vol_pre_cat = Rare		0.073 (0.153)
event_time = 2 × vol_pre_cat = Occasional		0.198 (0.154)
event_time = 2 × vol_pre_cat = Frequent		0.138 (0.153)
event_time = 3 × vol_pre_cat = Never		-0.105 (0.126)
event_time = 3 × vol_pre_cat = Rare		-0.143 (0.130)
event_time = 3 × vol_pre_cat = Occasional		-0.078 (0.131)
event_time = 3 × vol_pre_cat = Frequent		-0.127 (0.130)
event_time = 4 × vol_pre_cat = Never		-0.056 (0.144)
event_time = 4 × vol_pre_cat = Rare		-0.022 (0.147)
event_time = 4 × vol_pre_cat = Occasional		0.004 (0.149)
event_time = 4 × vol_pre_cat = Frequent		-0.070 (0.148)
event_time = 5 × vol_pre_cat = Never		0.257 (0.174)
event_time = 5 × vol_pre_cat = Rare		0.298+ (0.178)
event_time = 5 × vol_pre_cat = Occasional		0.250 (0.179)
event_time = 5 × vol_pre_cat = Frequent		0.288 (0.178)
Num.Obs.	46251	46251
R2	0.676	0.676
R2 Within	0.001	0.002

Standard errors clustered at the individual level. Fixed effects absorb time-invariant individual characteristics.

Appendix Table A4: Robustness models

Ordinal Robustness Models		
	Ordered logit: Binary volunteering	**Ordered logit: Volunteering frequency**
Bad|Poor	-2.380*** (0.095)	-2.380*** (0.095)
Poor|Satisfactory	-0.635*** (0.094)	-0.634*** (0.094)
Satisfactory|Good	1.132*** (0.094)	1.134*** (0.094)
Good|Very good	3.720*** (0.096)	3.723*** (0.096)
event_time_f-2	0.020 (0.038)	0.020 (0.038)
event_time_f0	0.029 (0.036)	0.029 (0.036)
event_time_f1	0.002 (0.038)	0.002 (0.038)
event_time_f2	-0.023 (0.039)	-0.023 (0.039)
event_time_f3	-0.010 (0.041)	-0.009 (0.041)
event_time_f4	-0.075+ (0.044)	-0.074+ (0.044)
event_time_f5	-0.074 (0.048)	-0.073 (0.048)
vol_pre_binaryAny volunteering	0.169*** (0.044)	
age_c	0.012*** (0.002)	0.012*** (0.002)

sexFemale	0.053** (0.017)	0.054** (0.017)
migbackYes	-0.089*** (0.027)	-0.088*** (0.027)
isced_clean_f2	0.056 (0.094)	0.055 (0.094)
isced_clean_f3	0.411*** (0.091)	0.412*** (0.091)
isced_clean_f4	0.647*** (0.099)	0.647*** (0.099)
isced_clean_f5	0.470*** (0.095)	0.468*** (0.095)
isced_clean_f6	0.707*** (0.092)	0.709*** (0.092)
isced_clean_fMissing	0.302 (0.229)	0.298 (0.229)
hinc_1k_c	0.115*** (0.005)	0.115*** (0.005)
event_time_f-2 × vol_pre_binaryAny volunteering	0.065 (0.065)	
event_time_f0 × vol_pre_binaryAny volunteering	0.051 (0.062)	
event_time_f1 × vol_pre_binaryAny volunteering	0.020 (0.065)	
event_time_f2 × vol_pre_binaryAny volunteering	0.041 (0.068)	
event_time_f3 × vol_pre_binaryAny volunteering	-0.027 (0.071)	
event_time_f4 × vol_pre_binaryAny volunteering	0.014 (0.076)	
event_time_f5 × vol_pre_binaryAny volunteering	-0.057 (0.084)	
vol_pre_catRare		0.122+ (0.067)

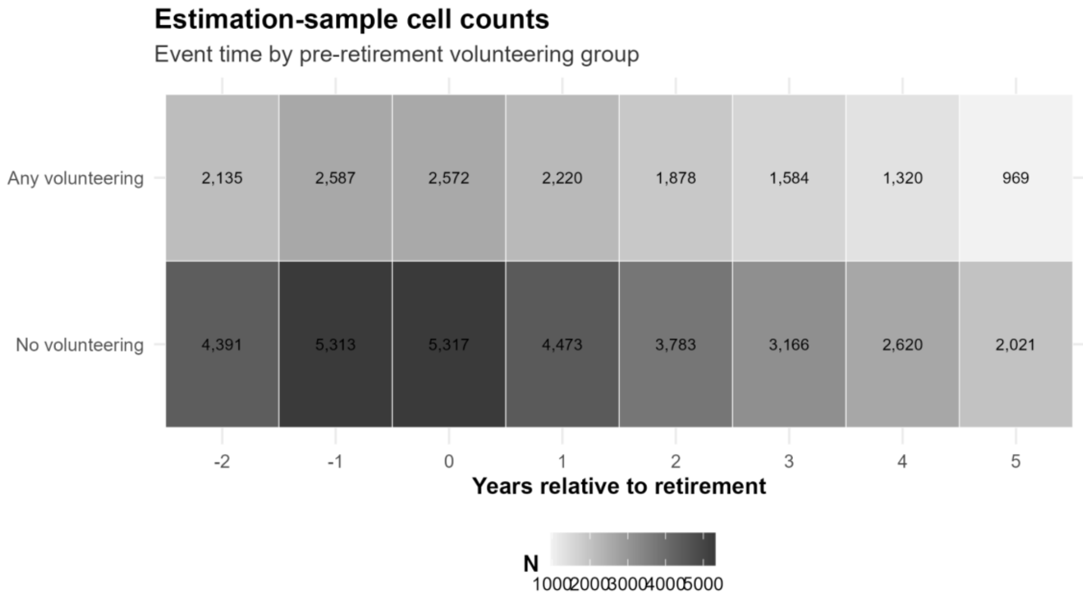
vol_pre_catOccasional		0.160* (0.071)
vol_pre_catFrequent		0.218** (0.067)
vol_pre_catVery frequent		0.300 (0.308)
event_time_f-2 × vol_pre_catRare		-0.005 (0.097)
event_time_f0 × vol_pre_catRare		0.045 (0.094)
event_time_f1 × vol_pre_catRare		0.032 (0.098)
event_time_f2 × vol_pre_catRare		-0.069 (0.101)
event_time_f3 × vol_pre_catRare		-0.091 (0.106)
event_time_f4 × vol_pre_catRare		-0.012 (0.112)
event_time_f5 × vol_pre_catRare		-0.059 (0.123)
event_time_f-2 × vol_pre_catOccasional		0.113 (0.105)
event_time_f0 × vol_pre_catOccasional		0.103 (0.100)
event_time_f1 × vol_pre_catOccasional		0.129 (0.104)
event_time_f2 × vol_pre_catOccasional		0.191+ (0.110)
event_time_f3 × vol_pre_catOccasional		0.087 (0.116)
event_time_f4 × vol_pre_catOccasional		0.129 (0.124)
event_time_f5 × vol_pre_catOccasional		-0.099 (0.138)

event_time_f-2 × vol_pre_catFrequent		0.135 (0.100)
event_time_f0 × vol_pre_catFrequent		0.043 (0.094)
event_time_f1 × vol_pre_catFrequent		-0.067 (0.097)
event_time_f2 × vol_pre_catFrequent		0.045 (0.103)
event_time_f3 × vol_pre_catFrequent		-0.063 (0.108)
event_time_f4 × vol_pre_catFrequent		-0.049 (0.115)
event_time_f5 × vol_pre_catFrequent		-0.001 (0.128)
event_time_f-2 × vol_pre_catVery frequent		-0.753 (0.491)
event_time_f0 × vol_pre_catVery frequent		-0.639 (0.436)
event_time_f1 × vol_pre_catVery frequent		-0.444 (0.461)
event_time_f2 × vol_pre_catVery frequent		-0.207 (0.472)
event_time_f3 × vol_pre_catVery frequent		0.359 (0.515)
event_time_f4 × vol_pre_catVery frequent		0.146 (0.555)
event_time_f5 × vol_pre_catVery frequent		-0.359 (0.573)
Num.Obs.	46349	46349
Ordered logit cumulative-link models. Model-based standard errors.		

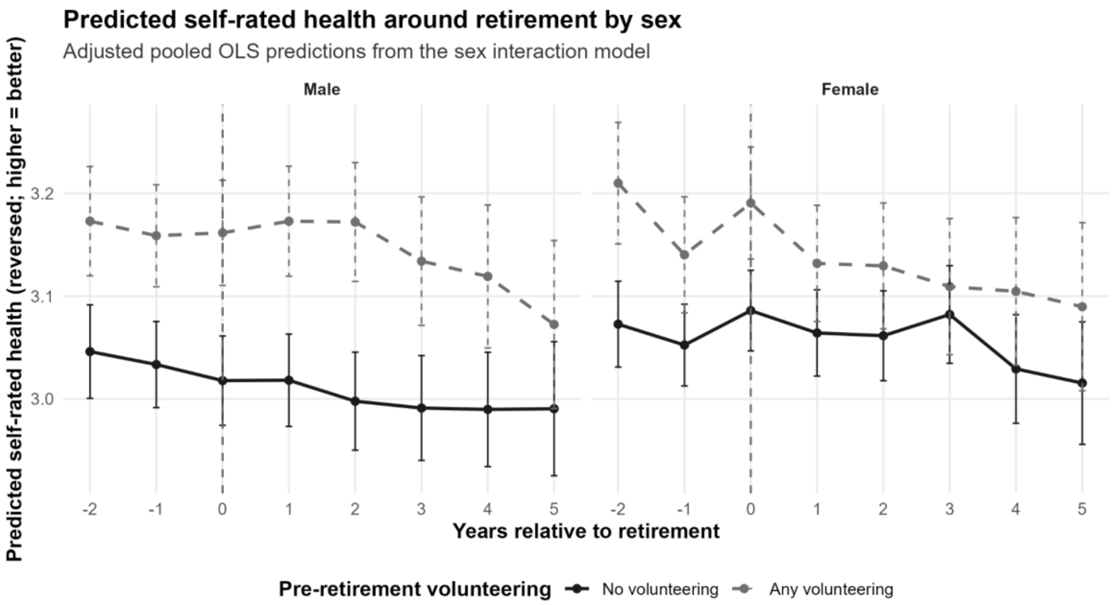
Appendix Table A5: Joint Wald tests

Joint Tests of Differential Health Trajectories				
test	**statistic**	**df1**	**df2**	**p_value**
Pooled OLS: all volunteering x event-time interactions	0.68	7.0	46323.00	0.69
Pooled OLS: post-retirement volunteering x event-time interactions	0.63	6.0	46323.00	0.70
FE model: all volunteering x event-time interactions	0.53	7.0	38138.00	0.81
FE model: post-retirement volunteering x event-time interactions	0.59	6.0	38138.00	0.74
Wald tests for the volunteering-by-event-time interaction terms.				

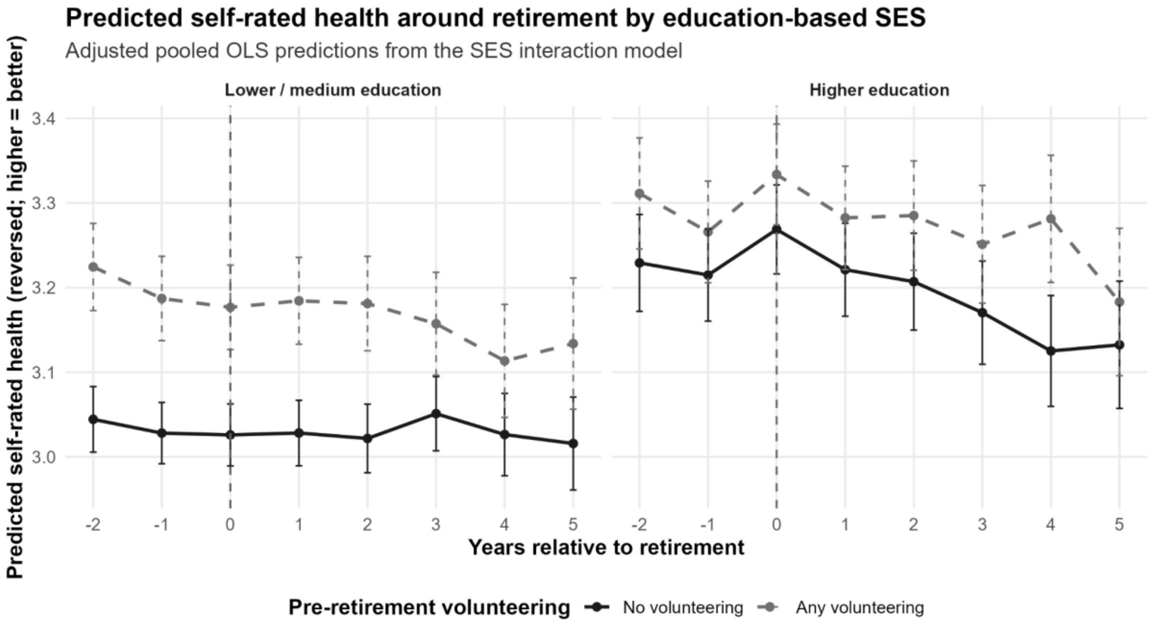
Appendix Figure A1: Estimation-sample cell-count heatmap



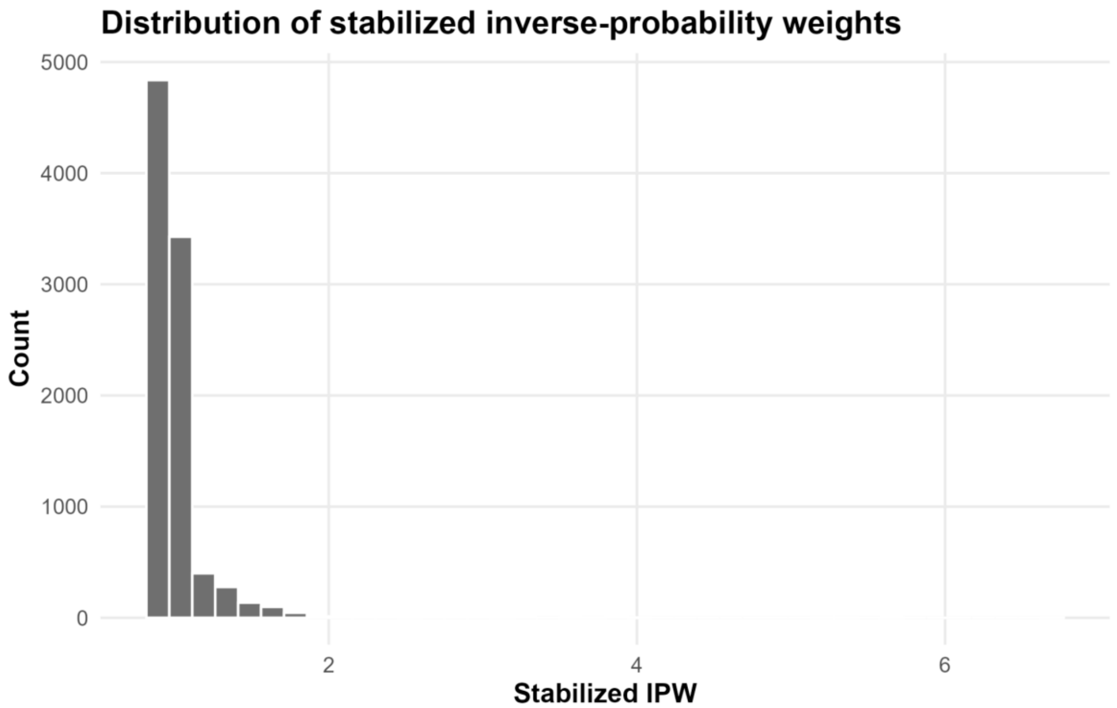
Appendix Figure A2: Heterogeneity by sex



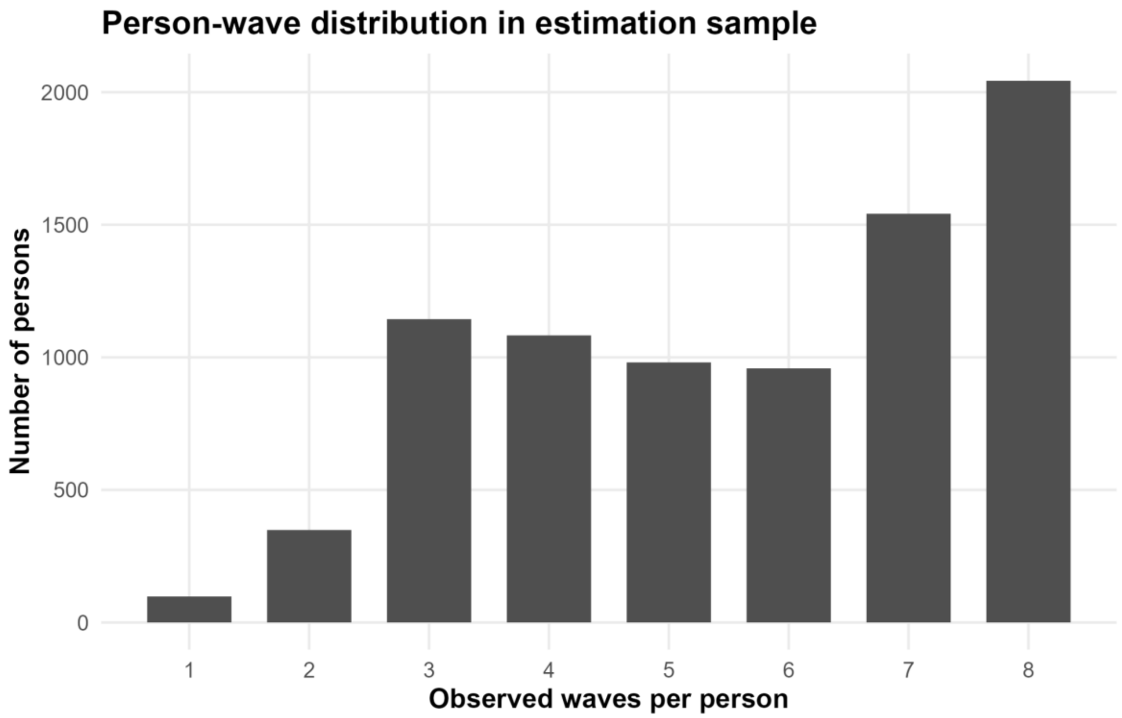
Appendix Figure A3: Heterogeneity by education-based SES



Appendix Figure A4: Distribution of stabilized IPW



Appendix Figure A5: Person-wave distribution



Data and Code Repository

The replication materials for this thesis are available in the following GitHub repository: https://github.com/SamuelPerezRestrepo/retirement-volunteering-health-soep

The repository contains the data-construction scripts, analysis code, and documentation required to reproduce the tables, figures, and main results reported in the thesis. Because the underlying SOEP microdata are subject to data protection restrictions, they cannot be redistributed through the repository. Researchers seeking to replicate the analysis must therefore request access to the SOEP data directly from DIW Berlin and follow the instructions provided in the repository README. Making the code and workflow publicly available strengthens the transparency and reproducibility of the study and facilitates future extensions of the analysis.